# ALGEBRAIC HOLOGRAPHIC HT SECTOR WITH VIRASORO BOUNDARY AND DERIVED CHIRAL CENTRE BULK

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Abstract. Three-dimensional Einstein gravity with negative cosmological constant is $SL(2,\mathbb{R}) \times SL(2,\mathbb{R})$ Chern–Simons (Achúcarro–Townsend, Witten); its boundary symmetry algebra is the Virasoro algebra at central charge $c = 3\ell/(2G)$ (Brown–Henneaux). We identify the *algebraic holographic HT sector* of this theory: the open Beilinson tower with boundary $\partial = \mathrm{Vir}_c$, bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) \simeq \mathbb{C}[[c]]$, and interaction the $\mathsf{SC}^{\mathrm{ch,top}}$-brace structure on the pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c), \mathrm{Vir}_c)$. The ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_c)$ on $\mathbb{C} \times \mathbb{R}$ is the twisting coalgebra; the derived chiral centre carries $E_3^{\mathrm{top}}$ structure via Sugawara topologisation (cohomological + weight-completed scope); the universal Maurer–Cartan element $\Theta_{\mathrm{Vir}_c}$ generates the shadow obstruction tower $\{S_r\}_{r\geq 2}$ recording the boundary-to-scalar trace of the algebraic sector.

The trace direction (operators $\to$ scalar) is well-defined and unconditional on the Virasoro Koszul locus: $\kappa(\mathrm{Vir}_c) = c/2$ is the Brown–Henneaux central charge; the quartic shadow $10/[c(5c+22)]$ matches the level-4 Kac determinant; the shadow free energies $F_g = \kappa\,\lambda_g^{\mathrm{FP}}$ reproduce the perturbative one-loop Hodge-integral expansion. The inverse direction (scalar $\to$ operator), in particular the construction of a compact dynamical-metric path integral from the algebraic sector, is the Vol I open frontier F12: it requires the BTZ / Cardy hypothesis package (modular invariance, vacuum dominance, saddle stability) together with the Maloney–Witten plus Manschot–Moore Farey-tail modular completion. Koszul duality $\mathrm{Vir}_c \leftrightarrow \mathrm{Vir}_{26-c}$ is the algebraic shadow of S-duality; the self-dual point sits at $c = 13$. The algebraic Page curve emerges from complementarity: $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$ forces saddle exchange at $t_P = 3S_{\mathrm{BH}}/13$, $c$-independent. The crossing is the structural shadow of the Page transition; the full information-theoretic content remains controlled by the island mechanism (Penington, AEMM). BTZ black holes are Maurer–Cartan elements of the gravitational convolution algebra; de Sitter entropy arises by analytic continuation $\kappa \to -|\kappa|$.

## Contents



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# 1. Three-dimensional gravity as Chern–Simons

Three-dimensional Einstein gravity with negative cosmological constant $\Lambda = -1/\ell^2$ has no propagating degrees of freedom. The Riemann tensor is determined algebraically by the Ricci tensor, which equals $\Lambda g_{\mu\nu}$; every solution is locally $\mathrm{AdS}_3$. The local triviality has an exact formulation: pure gravity in 2+1 dimensions is a gauge theory.

## 1.1. The Chern–Simons formulation.

**Theorem 1.1** (Achúcarro–Townsend [1], Witten [2]). *Three-dimensional gravity with cosmological constant $\Lambda = -1/\ell^2$ is classically equivalent to $SL(2,\mathbb{R}) \times SL(2,\mathbb{R})$ Chern–Simons theory at levels $k_\pm = \ell/(4G)$, where $G$ is Newton's constant. The gauge fields are*

$$A^a_\pm \;=\; \omega^a \pm \frac{e^a}{\ell}, \tag{1}$$

*where $e^a$ is the dreibein and $\omega^a$ the spin connection. Flatness $F_\pm = dA_\pm + A_\pm \wedge A_\pm = 0$ is equivalent to constant negative curvature plus torsion-freedom.*

*Proof.* The Einstein–Hilbert action with $\Lambda < 0$ in three dimensions reads

$$I_{\mathrm{EH}} \;=\; \frac{1}{16\pi G}\int \epsilon_{abc}\Big(e^a \wedge R^{bc} + \frac{1}{3\ell^2}\, e^a \wedge e^b \wedge e^c\Big),$$

where $R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}$ is the curvature two-form. The Chern–Simons functional for a gauge field $A$ on a three-manifold is $I_{\mathrm{CS}}[A] = (k/4\pi)\int \mathrm{tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$. Using the $\mathfrak{sl}_2$ trace $\mathrm{tr}(J_a J_b) = \frac{1}{2}\eta_{ab}$ and the decomposition (1):

$$I_{\mathrm{CS}}[A_+] - I_{\mathrm{CS}}[A_-] \;=\; \frac{k}{4\pi}\int \epsilon_{abc}\Big(2\, e^a \wedge R^{bc} + \frac{2}{3\ell^2}\, e^a \wedge e^b \wedge e^c\Big) \;=\; \frac{k}{2\pi}\, I_{\mathrm{EH}},$$

provided $k = \ell/(4G)$. The equations of motion $F_\pm = 0$ decompose into the torsion constraint $T^a = de^a + \epsilon^a{}_{bc}\,\omega^b \wedge e^c = 0$ and the Einstein equation $R^{ab} + (1/\ell^2) e^a \wedge e^b = 0$. □

*Remark* 1.2 (Scope: algebraic sector versus dynamical-metric path integral). The Achúcarro–Townsend–Witten equivalence holds at the classical level: both theories have the same equations of motion and the same moduli of flat connections. At the quantum level, the Chern–Simons path integral is a topological invariant (the Reshetikhin–Turaev functor), while a compact dynamical-metric gravitational path integral must also sum over topologies and saddle geometries.

The *algebraic holographic HT sector* of the gravitational theory is the open Beilinson tower $\partial = \mathrm{Vir}_c$, bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) \simeq \mathbb{C}[[c]]$, interaction the $\mathsf{SC}^{\mathrm{ch,top}}$-brace structure on $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c), \mathrm{Vir}_c)$. This sector is the level-3 / level-4 / level-5 Beilinson rungs of the gravitational theory; it is unconditional on the Virasoro Koszul locus (generic $c$).

The *trace direction* (operators $\longrightarrow$ scalar) descends from the algebraic sector to modular and partition-function shadows by trace + clutching on the open category; this is the content of Sections 4–5 and is unconditional. The *inverse direction* (scalar $\longrightarrow$ operator), in particular the construction of a compact dynamical-metric path integral from the algebraic sector, is the Vol I open frontier F12 and requires a distinct hypothesis package: the BTZ / Cardy hypotheses (modular invariance + vacuum dominance + saddle stability) together with the Maloney–Witten plus Manschot–Moore Farey-tail modular completion. The reverse direction is a separate conjectural inverse problem, stated in Section 8 and Section 12.

## 1.2. The Brown–Henneaux central charge.

The boundary dynamics is not trivial. Imposing asymptotically $\mathrm{AdS}_3$ boundary conditions at the conformal boundary $\partial(\mathrm{AdS}_3) \cong \mathbb{R} \times S^1$ produces a two-dimensional conformal field theory whose symmetry algebra is generated by the asymptotic charges.

**Theorem 1.3** (Brown–Henneaux [3]). *The asymptotic symmetry algebra of* $\mathrm{AdS}_3$ *gravity with* $\Lambda = -1/\ell^2$ *is two copies of the Virasoro algebra at central charge*

$$c \;=\; \frac{3\ell}{2G},$$

*equivalently* $c = 6k$ *with* $k = \ell/(4G)$ *the Chern–Simons level.*

*Proof.* In the Chern–Simons formulation, the boundary condition on the half-space $\mathbb{R}_{\geq 0} \times \mathbb{C}$ is the Fefferman–Graham gauge $A_z|_{t=0} = J_- + T(z)\,J_+$, where $J_\pm, J_0$ are the $\mathfrak{sl}_2$ generators with $[J_0, J_\pm] = \pm J_\pm$, $[J_+, J_-] = 2J_0$. The residual gauge transformations preserving this form are $g = \exp(\epsilon(z)\,J_+)$, acting by

$$\delta_\epsilon T \;=\; \epsilon'\,T + 2\epsilon\,T' + \frac{c}{12}\,\epsilon''',$$

which is the Virasoro coadjoint action. The Chern–Simons symplectic form induces the Poisson bracket

$$\{T(z),\, T(w)\}_{\mathrm{PB}} \;=\; \frac{c}{2}\,\delta'''(z-w) + 2T(w)\,\delta'(z-w) + T'(w)\,\delta(z-w),$$

which is the defining relation of $\mathrm{Vir}_c$. The central charge is $c = 6k = 3\ell/(2G)$. $\square$

1.3. **The input datum.** The entire construction of this paper flows from the Virasoro $\lambda$-bracket, which encodes the OPE of the boundary stress tensor in algebraic form:

$$\{T_\lambda T\} \;=\; \partial T + 2T\lambda + \frac{c}{12}\,\lambda^3. \tag{2}$$

The three terms encode, respectively: the translation $T_{(0)}T = \partial T$ (simple pole), the dilatation $T_{(1)}T = 2T$ (double pole), and the central extension $T_{(3)}T = c/2$ (quartic pole). In the OPE formalism:

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{3}$$

The triple pole $T_{(2)}T = 0$ vanishes: the coefficient of $\lambda^2$ in the $\lambda$-bracket is zero, forced by skew-symmetry. The quartic pole is the single datum from which the entire gravitational structure follows.

*Remark* 1.4 (Physical parameters). The central charge $c = 3\ell/(2G)$ is the only free parameter of 3d gravity with $\Lambda < 0$. In Planck units $\ell = 1$, it is $c = 3/(2G)$: the inverse coupling constant. The semiclassical limit $c \to \infty$ is $G \to 0$. The curvature $\kappa(\mathrm{Vir}_c) = c/2$ records the same information; the relation $c = 2\kappa$ persists throughout.

1.4. **The holomorphic-topological structure.** The Chern–Simons theory on $\mathbb{R}_{\geq 0} \times \mathbb{C}$ has a natural factorisation into holomorphic and topological directions: the $\mathbb{C}$-direction carries the holomorphic OPE structure (the vertex algebra of the boundary), while the $\mathbb{R}_{\geq 0}$-direction carries the topological ordering (the associative structure of the bar complex). The product $\mathbb{C} \times \mathbb{R}$ is the habitat of the $E_1$-chiral bar complex $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_c)$, which is a dg coalgebra over the Koszul dual cooperad $(\mathrm{ChirAss})^!$: the bar differential uses the holomorphic OPE data from $\mathrm{FM}_k(\mathbb{C})$, while the deconcatenation coproduct uses the ordered topological structure from $\mathrm{Conf}_k^<(\mathbb{R})$.

The $\mathsf{SC}^{\mathrm{ch,top}}$ structure on the chiral derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$ (see Section 3) becomes $E_3$-topological via Sugawara topologisation at non-critical level: the Sugawara stress tensor $T(z)$ satisfies $T = \{Q_{\mathrm{CS}}, G'\}$ in the BRST complex, so $\mathbb{C}$-translations are $Q$-exact, the complex structure on $\mathbb{C}$ becomes irrelevant in cohomology, and the two $\mathsf{SC}^{\mathrm{ch,top}}$ colours collapse. The result is an $E_3$-topological algebra: $E_2^{\mathrm{top}} \times E_1^{\mathrm{top}}$ via Dunn additivity, independent of the complex structure on $\mathbb{C}$. This is proved for the affine Kac–Moody algebra $V_k(\mathfrak{sl}_2)$ at non-critical level $k \neq -2$; for the Virasoro algebra obtained by Drinfeld–Sokolov reduction, the topologisation is cohomological and the chain-level structure is conditional on $A_\infty$ coherence.

## 2. The Virasoro bar complex

The ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_c) = T^c(s^{-1}\overline{\mathrm{Vir}}_c)$ is an $E_1$-chiral coassociative coalgebra: the bar differential $d_{\bar{B}}$ comes from the OPE residues at the boundary strata of $\mathrm{FM}_k(\mathbb{C})$, and the deconcatenation coproduct $\Delta\colon \bar{B} \to \bar{B} \otimes \bar{B}$ is the cofree tensor coalgebra structure on $T^c(s^{-1}\overline{\mathrm{Vir}}_c)$. The bar complex is a single-coloured $E_1$ coalgebra; it is not an $\mathsf{SC}^{\mathrm{ch,top}}$-coalgebra (the Swiss-cheese structure emerges on the derived centre pair $(C^\bullet_{\mathrm{ch}}(\mathrm{Vir}_c, \mathrm{Vir}_c), \mathrm{Vir}_c)$, not on $\bar{B}$).

2.1. **The $A_\infty$ operations.** The $\lambda$-bracket (2) determines the binary $A_\infty$ operation $m_2$ directly. The quartic pole forces $m_2$ to be non-associative; the Stasheff identity then forces all higher operations $m_3, m_4, \ldots$ in perpetuity: the tower is infinite, and $\mathrm{Vir}_c$ is class $\mathbf{M}$.

**Definition 2.1** (The binary operation)**.** The binary $A_\infty$ operation for $\mathrm{Vir}_c$ is the $\lambda$-bracket itself:

$$m_2(T, T; \lambda) \;=\; \{T_\lambda T\} \;=\; \frac{c}{12}\lambda^3 + 2T\lambda + \partial T. \tag{4}$$

**Proposition 2.2** (The Virasoro associator)**.** *The associator* $A_3 = m_2 \circ (\mathrm{id} \otimes m_2) - m_2 \circ (m_2 \otimes \mathrm{id})$ *is*

$$A_3(T, T, T;\, \lambda_{12}, \lambda_{23}) \;=\; -\partial^2 T - (2\lambda_{12} + 3\lambda_{23})\,\partial T - 2\lambda_{23}(2\lambda_{12} + \lambda_{23})\,T - \frac{c}{12}\lambda_{23}^3(2\lambda_{12} + \lambda_{23}). \tag{5}$$

*In particular $A_3 \neq 0$ whenever $c \neq 0$ or the field-dependent terms are non-trivial: the $\lambda$-bracket is not associative. For affine Kac–Moody algebras (where the quartic pole vanishes), $A_3 = 0$ and $m_2$ is strictly associative.*

*Proof.* The PVA Jacobi identity gives $A_3(\ell, \mu) = -\{T_\mu\{T_\ell T\}\}$. Direct computation using sesquilinearity of the $\lambda$-bracket: the inner bracket $\{T_\ell T\} = \partial T + 2T\ell + (c/12)\ell^3$ is substituted into the outer bracket using $\{T_\mu(\partial T)\} = -(\mu + \partial)\{T_\mu T\}$ (the left sesquilinearity identity) and $\{T_\mu T\} = \partial T + 2T\mu + (c/12)\mu^3$. The scalar term $\{T_\mu (c/12)\ell^3\} = 0$ (the bracket of a constant vanishes). Collecting terms yields (5). □

The Stasheff identity $m_3 = -A_3$ then gives the ternary operation.

**Proposition 2.3** (The ternary operation)**.** *The ternary $A_\infty$ operation for* $\mathrm{Vir}_c$ *is*

$$m_3(T, T, T;\, \lambda_{12}, \lambda_{23}) \;=\; \partial^2 T + (2\lambda_{12} + 3\lambda_{23})\,\partial T + 2\lambda_{23}(2\lambda_{12} + \lambda_{23})\,T + \frac{c}{12}\lambda_{23}^3(2\lambda_{12} + \lambda_{23}). \tag{6}$$

*The four terms encode: the second-order geometric correction $(\partial^2 T)$, the first-order response $((2\lambda_{12} + 3\lambda_{23})\partial T)$, the stress-tensor backreaction $(2\lambda_{23}(2\lambda_{12} + \lambda_{23})T)$, and the three-graviton coupling $((c/12)\lambda_{23}^3(2\lambda_{12} + \lambda_{23}))$.*

**Proposition 2.4** (The quaternary operation)**.** *The quaternary $A_\infty$ operation, determined by the degree-4 Stasheff identity on $\mathrm{FM}_4(\mathbb{C}) \cong K_4$ (five compositions: two from $(r, s) = (2, 3)$, three from $(r, s) = (3, 2)$), is*

$$\begin{aligned} m_4(T^4; \lambda_{12}, \lambda_{23}, \lambda_{34}) &= (\lambda_{23} + 2\lambda_{34})\,\partial^2 T \\ &\quad + (5\lambda_{12}^2 - 4\lambda_{12}\lambda_{23} + 3\lambda_{12}\lambda_{34} - 5\lambda_{23}^2 + 10\lambda_{23}\lambda_{34} + 4\lambda_{34}^2)\,\partial T \\ &\quad + \textit{(degree-3 polynomial)} \cdot T \\ &\quad + \frac{c}{12}\,Q_4(\lambda_{12}, \lambda_{23}, \lambda_{34}), \end{aligned}$$

*where $Q_4$ is a homogeneous polynomial of degree 5. At the symmetric point $\lambda_{12} = \lambda_{23} = \lambda_{34} = \lambda$:*

$$m_4(T^4;\, \lambda, \lambda, \lambda) \;=\; 3\lambda\,\partial^2 T + 13\lambda^2\,\partial T + 14\lambda^3\,T + \frac{7c}{12}\lambda^5. \tag{7}$$

*The leading term is $\partial^2 T$ (at spectral depth 2), not $\partial^3 T$ (depth 1): the depth-1 coefficient vanishes by palindromic cancellation among the Stasheff compositions.*

*Proof.* Solve the degree-4 Stasheff relation using the sesquilinearity of $m_2$ (4) and $m_3$ (6): the first-slot rule $m_3(\partial^j T, \dots) = (-\lambda_{12})^j m_3(T, \dots)$ and the last-slot rule $m_3(\dots, \partial^j T) = (\lambda_{23} + \partial)^j m_3(\dots, T)$ reduce every composition to a polynomial in $\lambda_{12}, \lambda_{23}, \lambda_{34}$. The symmetric-point evaluation (7) is verified by direct computation. □

2.2. **The classical $r$-matrix.** The classical $r$-matrix is obtained from the $\lambda$-bracket by the collision residue construction: the bar kernel $d\log(z-w)$ absorbs one pole, so the $r$-matrix has poles one order lower than the OPE (the $d\log$ absorption principle).

**Proposition 2.5** (Virasoro collision residue). *The collision residue of the Virasoro $\lambda$-bracket is*

$$r^{\mathrm{coll}}(z) \;=\; \frac{c/2}{z^3} + \frac{2T}{z}, \tag{8}$$

*a cubic-plus-simple rational function. The OPE has a quartic pole; the $d\log$ absorption reduces the order by one. The Laplace kernel (without $d\log$ absorption) is $r^L(z) = (c/2)/z^4 + 2T/z^2 + \partial T/z$.*

*Remark* 2.6 (Convention). Two conventions coexist in the literature. The collision residue (8) absorbs $d\log(z)$ (one pole fewer than OPE). The Laplace kernel $r^L(z)$ retains the full OPE poles. We use the collision residue throughout, following the bar-complex convention where the propagator is $d\log E(z,w)$ (weight 1 always). At $c = 0$ the scalar part of $r^{\mathrm{coll}}$ vanishes: the abelian limit, consistent with $\kappa(\mathrm{Vir}_0) = 0$.

**Proposition 2.7** (Yang–Baxter equation). *The CYBE*

$$[r_{12}(z_{12}),\, r_{13}(z_{13})] + [r_{12}(z_{12}),\, r_{23}(z_{23})] + [r_{13}(z_{13}),\, r_{23}(z_{23})] = 0$$

*holds for the collision residue* (8). *It is the projection to the binary collision stratum of the Arnold relation $\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{13} + \omega_{13} \wedge \omega_{12} = 0$ on $\mathrm{FM}_3(\mathbb{C})$, with $\omega_{ij} = d\log(z_i - z_j)$.*

2.3. **The $c$-linearity theorem.** A structural pattern persists across all computed degrees $m_2$ through $m_5$: the central charge enters each operation through a single scalar contact term, while all stress-tensor-dependent coefficients are $c$-independent. This separates the universal gravitational backreaction from the coupling-dependent scattering amplitude.

**Theorem 2.8** ($c$-linearity). *For all $k \geq 2$, the scalar part of $m_k(T^{\otimes k}; \lambda_1, \dots, \lambda_{k-1})$ is **linear** in $c$:*

$$m_k\big|_{\mathrm{scalar}} \;=\; \frac{c}{12}\, P_k(\lambda_1, \dots, \lambda_{k-1}),$$

*where $P_k$ is a homogeneous polynomial of degree $k+1$. All $T$-dependent coefficients (the polynomials multiplying $\partial^j T$ for $0 \leq j \leq k-2$) are independent of $c$.*

*Proof.* The central charge enters the $\lambda$-bracket (2) as a scalar contact term $(c/12)\lambda^3$. Scalars are annihilated by the bracket in both slots: $\{1{}_\lambda T\} = 0 = \{T{}_\lambda 1\}$ (PVA vacuum axiom). Induction on the Stasheff relations: each $m_k$ is a sum of compositions of lower $m_j$, and the only $c$-dependent input is the scalar part of $m_2$. Since scalars are killed by subsequent brackets, $c$ propagates linearly through the recursion. The degree count follows from $\deg P_2 = 3$ and the Stasheff composition raising degree by 2 at each step. □

*Remark* 2.9 (Gravitational interpretation). The theorem states that graviton self-interaction at every degree is controlled by a single coupling: $c/12 = \kappa/6$, where $\kappa(\mathrm{Vir}_c) = c/2$. The stress-tensor backreaction ($T, \partial T, \dots, \partial^{k-2}T$ terms) is universal, independent of the cosmological constant. The pure-graviton scattering amplitude is proportional to $c/12$: Newton's constant enters once.

### 2.4. **Infinite tower and class M.**

**Theorem 2.10** (Non-truncation). *The $A_\infty$ tower of $\mathrm{Vir}_c$ does not truncate: $m_k \neq 0$ for all $k \geq 3$ and generic $c$. More precisely:*

(i) *The quartic OPE pole forces $A_3 \neq 0$, hence $m_3 = -A_3 \neq 0$.*
(ii) *The Stasheff identity at each subsequent degree forces $m_k \neq 0$ from the non-vanishing of the wheel integral on the associahedron $K_k$.*
(iii) *At $c = 0$, the scalar terms vanish but the field-dependent terms ($\partial^j T$ coefficients) survive unchanged: the tower remains infinite even without central charge.*

*The discriminant $\Delta = 8\kappa S_4 = 8(c/2) \cdot S_4 \neq 0$ for generic $c$, confirming class* **M***: infinite shadow depth.*

*Proof.* Part (i): the associator (5) contains the field-independent term $-\partial^2 T$, which is nonzero regardless of $c$, and the $c$-dependent term $-(c/12)\lambda_{23}^3(2\lambda_{12} + \lambda_{23})$, which is nonzero for $c \neq 0$.

Part (ii): the degree-$k$ Stasheff identity is $\sum_{i+j=k+1}(-1)^{i(j-1)} m_i \circ m_j = 0$. The compositions involve the previously established nonzero operations $m_2, \ldots, m_{k-1}$, and the obstruction at degree $k$ is nonzero by the wheel integral computation: the Hamiltonian cycle on the complete graph $K_k$ with $k-1$ cubic vertices has nonzero residue from the quartic OPE pole.

Part (iii): by the $c$-linearity theorem (Theorem 2.8), setting $c = 0$ kills only the scalar contact terms. The field-dependent coefficients are $c$-independent and nonzero (the self-coupling $T_{(1)}T = 2T$ drives the induction). □

*Remark* 2.11 (The Drinfeld–Sokolov mechanism). The infinite $A_\infty$ tower is manufactured by Drinfeld–Sokolov reduction from the *finite* Kac–Moody tower. The principal DS reduction of $V_k(\mathfrak{sl}_2)$ produces $\mathrm{Vir}_c$ with $c = 1 - 6(k+1)^2/(k+2)$, transporting the complexity class from **L** (affine: $m_k = 0$ for $k \geq 3$) to **M** (Virasoro: all $m_k \neq 0$). The mechanism: the DS stress tensor $T^{\mathrm{DS}} = J^f + (J^h)^2/(4(k+2)) + \frac{1}{2}\partial J^h$ contains a quadratic Sugawara term whose double Wick self-contraction generates the quartic OPE pole, absent in the original affine OPE. The manufactured quartic pole makes $m_2$ non-associative, which seeds the entire infinite tower.

### 2.5. **The deconcatenation coproduct.**
The bar complex $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_c) = T^c(s^{-1}\overline{\mathrm{Vir}}_c)$ carries the deconcatenation coproduct

$$\Delta[s^{-1}a_1|\cdots|s^{-1}a_n] = \sum_{i=0}^{n}[s^{-1}a_1|\cdots|s^{-1}a_i] \otimes [s^{-1}a_{i+1}|\cdots|s^{-1}a_n],$$

which is the cofree coassociative coalgebra structure. Together with the bar differential $d_{\bar{B}}$, this makes $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_c)$ an $E_1$ dg coassociative coalgebra. The coproduct is structural (it exists for any bar complex); the gravitational content is in the differential $d_{\bar{B}}$, which encodes the $A_\infty$ operations via the OPE residues.

The Drinfeld–Sokolov transfer theorem shows that the *spectral* coproduct $\Delta_z$ (the deformation of the deconcatenation by the spectral parameter) is primitive on $\mathrm{Vir}_c$: $\Delta_z(T) = \tau_z(T) \otimes 1 + 1 \otimes T$, with no corrections at any order. The mechanism is a ghost-number obstruction: the homotopy $h$ of the DS strong deformation retract has ghost degree $-1$, and every tree in the HPL expansion of the transferred coproduct at degree $n \geq 2$ contains an uncompensated $h$-insertion that deposits the intermediate state in a ghost sector invisible to the projection. Gravitons do not split.

## 3. The derived chiral centre and the gravitational comparison

The bar complex computes the boundary data. The level-3 closed-sector object is the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) = C^\bullet_{\mathrm{ch}}(\mathrm{Vir}_c, \mathrm{Vir}_c)$: the chiral Hochschild cochain complex of the boundary algebra, defined via the chiral endomorphism operad $\mathrm{End}^{\mathrm{ch}}_{\mathrm{Vir}_c}$ with spectral parameters from $\mathrm{FM}_k(\mathbb{C})$. A gravitational bulk is obtained from it only after the Virasoro open–closed comparison map has been supplied.

### 3.1. Chiral Hochschild cohomology of the Virasoro algebra.

**Theorem 3.1** (Chiral Hochschild cohomology of $\mathrm{Vir}_c$). ***Type signature.*** *Quadrant: Open. Presentation: weight-completed / pro-object ambient (Class* **M** *chain-level on the original direct-sum complex remains conditional; see Frontier F3). Level: 3 (derived chiral centre as universal closed sector). Hypothesis package: generic c (Koszul locus); finite-dimensional graded augmented Virasoro vacuum module; Hochschild–Serre spectral-sequence convergence in the completed ambient.*

*For generic c, the chiral Hochschild cohomology of* $\mathrm{Vir}_c$ *is concentrated in degrees* $\{0, 1, 2\}$ *with Hilbert series* $P(t) = 1 + t^2$*:*

$$\begin{aligned}
\mathrm{ChirHoch}^0(\mathrm{Vir}_c) &= \mathbb{C},\\
\mathrm{ChirHoch}^1(\mathrm{Vir}_c) &= 0,\\
\mathrm{ChirHoch}^2(\mathrm{Vir}_c) &= \mathbb{C}\cdot\Theta_c,\\
\mathrm{ChirHoch}^n(\mathrm{Vir}_c) &= 0 \qquad \textit{for } n \geq 3 \textit{ and generic } c,
\end{aligned}$$

*where* $\Theta_c \in \mathrm{ChirHoch}^2$ *is the Gel'fand–Fuchs 2-cocycle recording* $\partial_c\{T_\lambda T\} = \lambda^3/12$.

*Proof.* The Hochschild–Serre spectral sequence for $\mathrm{Vir}_c$ collapses at $E_2$ for generic $c$. In degree 0: the centre of $\mathrm{Vir}_c$ is $\mathbb{C}$ (the Virasoro algebra is simple for generic $c$). In degree 1: every derivation is inner (the Virasoro algebra has $H^1(\mathrm{Vir}_c, \mathrm{Vir}_c) = 0$ for $c \neq 0$, since the only derivation $D$ satisfying $D[L_m, L_n] = [DL_m, L_n] + [L_m, DL_n]$ is $D = \mathrm{ad}(v)$ for some $v$). In degree 2: the Gel'fand–Fuchs computation gives $H^2 = \mathbb{C}$ for generic $c$, generated by the central extension cocycle $\Theta_c(L_m, L_n) = \frac{1}{12}(m^3 - m)\delta_{m+n,0}$. In degree $\geq 3$: vanishing by spectral sequence degeneration at generic $c$ (the singular locus in the $c$-plane where higher cohomology appears is discrete). □

**Corollary 3.2** (The bulk of 3d gravity). *The derived chiral centre of* $\mathrm{Vir}_c$ *is*

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) \;\simeq\; \mathbb{C}[[c]].$$

*The unique bulk observable is the central charge parameter: 3d gravity has no local degrees of freedom.*

*Proof.* By Theorem 3.1, the Hilbert series is $P(t) = 1 + t^2$. The degree-0 component $\mathbb{C}$ is the identity; the degree-2 component $\mathbb{C}\cdot\Theta_c$ is the deformation parameter $c$. The formal power series ring $\mathbb{C}[[c]]$ is the completed algebra of functions on the one-dimensional formal moduli space parametrised by the central charge. □

**3.2. The Swiss-cheese structure and $E_3$-topologisation.** The chiral derived centre carries a natural $\mathsf{SC}^{\mathrm{ch,top}}$ structure: the pair $(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c), \mathrm{Vir}_c)$ is a two-coloured datum where the bulk (closed colour) acts on the boundary (open colour). This is the algebraic encoding of the bulk-to-boundary map in $\mathrm{AdS}_3/\mathrm{CFT}_2$.

**Theorem 3.3** ($E_3$-topologisation for Virasoro). ***Type signature.*** *Quadrant: Open. Presentation: cohomological + weight-completed (chain-level on the original direct-sum complex is Frontier F3, conditional on the Felder BRST Mittag–Leffler tower). Level: 3 (derived chiral centre). Hypothesis package: non-critical level* $k \neq -2$ *on the affine side; Sugawara primitive* $G_{\mathrm{Sug}}$ *with* $[Q_{\mathrm{tot}}, G_{\mathrm{Sug}}] = T_{\mathrm{Sug}}$ *on cohomology;* $A_\infty$ *coherence of the topologisation primitives in the chain-level lift (open at Class* **M**).

*For* $\mathrm{Vir}_c$ *obtained by Drinfeld–Sokolov reduction from* $V_k(\mathfrak{sl}_2)$ *at non-critical level* $k \neq -2$ *(the critical level* $k = -h^\vee = -2$ *maps to* $c \to \infty$ *under* $c = 1 - 6(k+1)^2/(k+2)$*; the condition excludes this singularity), the* $\mathsf{SC}^{\mathrm{ch,top}}$ *structure on the derived centre pair promotes to an* $E_3$*-topological structure on*

$H^\bullet(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c), Q)$. *The mechanism: the Sugawara stress tensor satisfies* $T(z) = \{Q_{\mathrm{CS}}, G'(z)\}$ *in the BRST complex, so* $\mathbb{C}$*-translations are* $Q$*-exact, the two* $\mathsf{SC}^{\mathrm{ch,top}}$ *colours collapse in cohomology, and* $E_2^{\mathrm{hol}} \times E_1^{\mathrm{top}} \to E_2^{\mathrm{top}} \times E_1^{\mathrm{top}} = E_3^{\mathrm{top}}$ *via Dunn additivity.*

*Remark* 3.4 (Scope). The $E_3$-topological structure is proved cohomologically: it holds on $H^\bullet(Z^{\mathrm{der}}, Q)$, not necessarily at the cochain level. For class **M** algebras (Virasoro), the chain-level $E_3$ is conditional on $A_\infty$ coherence of the topologisation. The cohomological statement suffices for the genus-tower computations of Section 4. The proof is for affine Kac–Moody $V_k(\mathfrak{sl}_2)$ at non-critical level, transported to Virasoro by the DS intertwining theorem; general chiral algebras with conformal vector are conjectural.

3.3. **The bulk-to-boundary map.** The bulk-to-boundary map sends the degree-2 chiral Hochschild class $\Theta_c \in \mathrm{ChirHoch}^2$ to the stress tensor $T \in \mathrm{Vir}_c$ via the conformal Ward identity:

$$\beta_B(\partial_c)(z_0) \cdot T(w) \;=\; \frac{T(w)}{(z_0 - w)^2} + \frac{\partial T(w)}{z_0 - w}. \tag{9}$$

This completes the *Koszul triangle*: bulk $\xrightarrow{\beta_B}$ boundary $\xrightarrow{\bar{B}}$ lines $\xrightarrow{Z^{\mathrm{der}}}$ bulk. The bar complex is the universal coalgebra from which three functors extract the three vertices:

(i) $\Omega(\bar{B}) \simeq \mathrm{Vir}_c$ (cobar recovers the boundary; bar-cobar adjunction),
(ii) $\Omega(D_{\mathrm{Ran}}(\bar{B})) \simeq \mathrm{Vir}_{26-c}$ (Verdier dual followed by cobar gives the Koszul dual; see Section 5),
(iii) $\mathrm{RHom}(\Omega(\bar{B}), \mathrm{Vir}_c) \simeq \mathbb{C}[[c]]$ (chiral Hochschild cochains give the universal closed sector; a physical bulk requires the open–closed comparison).

## 4. The shadow tower as algebraic-sector trace

The universal Maurer–Cartan element $\Theta_{\mathrm{Vir}_c}$ in the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathrm{Vir}_c}$ satisfies the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ and encodes the algebraic-sector trace of the perturbative expansion of 3d gravity (operator $\to$ scalar trace direction; level-4 $\to$ level-5 of the Beilinson tower). Its projections to successive degrees constitute the *shadow obstruction tower*, a sequence of invariants $\{S_r\}_{r\geq 2}$ that record the perturbative corrections to the semiclassical gravitational dictionary at the level of the algebraic sector. The inverse construction (from the $\{S_r\}$ to a compact dynamical-metric path integral) is the named open obstruction F12; here we compute the trace.

### 4.1. The shadow invariants.

**Definition 4.1** (Shadow tower of $\mathrm{Vir}_c$). The shadow obstruction tower is the sequence $S_2, S_3, S_4, \ldots$ of projections of the MC element $\Theta_{\mathrm{Vir}_c}$ to successive degrees in the convolution algebra. The first four named invariants are:

(i) $S_2 = \kappa(\mathrm{Vir}_c) = c/2$: the curvature of the bar complex (UNIFORM-WEIGHT).
(ii) $S_3 = \mathfrak{C}(\mathrm{Vir}_c) = -c$: the cubic shadow, encoding the Lie bracket structure of the convolution algebra.
(iii) $S_4$: the quartic shadow, which determines the quartic contact invariant

$$\mathfrak{Q}^{\mathrm{contact}}_{\mathrm{Vir}} \;=\; \frac{10}{c(5c+22)}. \tag{10}$$

The numerator 10 arises from the Shapovalov inner product at level 4; the denominator $c(5c+22)$ is the Kac determinant. Poles at $c = 0$ ($G = \infty$) and $c = -22/5$ (Lee–Yang) mark reorganisation points of the tower.

(iv) $S_r$ for $r \geq 5$: the higher shadow obstructions, each determined by the degree-$r$ Stasheff identity and the Kac determinant at levels $\leq r$.

### 4.2. **The discriminant and class M.**

**Theorem 4.2** (Discriminant). *The discriminant of* $\mathrm{Vir}_c$ *is*

$$\Delta(\mathrm{Vir}_c) \;=\; 8\,\kappa\, S_4 \;=\; 8\cdot\frac{c}{2}\cdot S_4 \;\neq\; 0 \tag{11}$$

*for generic c (in particular for all $c > 0$). The discriminant is linear in κ, not quadratic. Since $\Delta \neq 0$, the shadow tower does not terminate:* $\mathrm{Vir}_c$ *is class* **M** *with infinite shadow depth $r_{\max} = \infty$.*

*Proof.* The quartic OPE pole forces $S_4 \neq 0$ (Proposition 2.4): the scalar contact term $(7c/12)\lambda^5$ at the symmetric point is proportional to $c$ and nonzero for $c \neq 0$. Combined with $\kappa = c/2 \neq 0$ for $c \neq 0$: $\Delta = 8(c/2)S_4 \neq 0$. When $\Delta \neq 0$, the recursion governing the shadow tower propagates to all genera, giving $r_{\max} = \infty$. □

### 4.3. **Gravitational interpretation of the shadow degrees.**

Each shadow degree has a direct gravitational reading. The table:

| **Shadow** | **Algebraic content** | **Gravitational counterpart** |
|---|---|---|
| $\kappa = c/2$ | bar complex curvature | Brown–Henneaux central charge; one-loop Weyl anomaly |
| $\mathfrak{C} = -c$ | cubic Lie bracket | cubic graviton vertex $\propto \ell/(16\pi G)$ |
| $\mathfrak{Q} = \frac{10}{c(5c+22)}$ | quartic contact | four-graviton amplitude; genus-2 contact correction |
| $S_r, r \geq 5$ | degree-$r$ obstruction | $r$-graviton vertex; higher-loop correction |

### 4.4. **The genus tower.**

The curvature $\kappa = \kappa(\mathrm{Vir}_c) = c/2$ controls the scalar genus shadow of the gravitational partition function through the fiberwise differential $d_{\text{fib}}^2 = \kappa \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$.

**Theorem 4.3** (The intrinsic genus tower). *The intrinsic genus-g free energy of* $\mathrm{Vir}_c$ *is (UNIFORM-WEIGHT, all $g \geq 1$):*

$$F_g^{\text{intr}}(\mathrm{Vir}_c) \;=\; \kappa(\mathrm{Vir}_c)\cdot\lambda_g^{\text{FP}}, \qquad \lambda_g^{\text{FP}} \;=\; \int_{\overline{\mathcal{M}}_g}\lambda_g \;=\; \frac{2^{2g-1}-1}{2^{2g-1}}\cdot\frac{|B_{2g}|}{(2g)!} \;>\; 0.$$

*The generating function is the $\hat{A}$-genus:*

$$\sum_{g\geq 1} F_g^{\text{intr}}\, x^{2g} \;=\; \frac{c}{2}\big(\hat{A}(ix)-1\big), \qquad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}.$$

*The first two terms:*

$$F_1^{\text{intr}} = \frac{\kappa}{24} = \frac{c}{48},$$
$$F_2^{\text{intr}} = \frac{7\kappa}{5760} = \frac{7c}{11520}.$$

*Proof.* The genus-$g$ free energy is $F_g^{\text{intr}} = \kappa\cdot\int_{\overline{\mathcal{M}}_g}\lambda_g$ by the scalar genus tower theorem applied to the single-generator algebra $\mathrm{Vir}_c$ (uniform weight: one generating field of conformal weight 2). The Hodge integrals $\lambda_g^{\text{FP}}$ are the Faber–Pandharipande values, whose generating function is $\sum_{g\geq 1}\lambda_g^{\text{FP}}x^{2g} = \hat{A}(ix) - 1 = (x/2)/\sin(x/2) - 1$ by the Mittag-Leffler expansion $(x/2)/\sin(x/2) = 1 + 2\sum_{g\geq 1}\frac{(2^{2g-1}-1)|B_{2g}|}{(2g)!}\,x^{2g}$. The substitution $x \mapsto ix$ converts $\sinh(ix/2) = i\sin(x/2)$, producing the $\hat{A}(ix)$ form with all positive coefficients (the Wick rotation from Lorentzian to Euclidean signature).

For $F_1^{\mathrm{intr}}$: $\lambda_1^{\mathrm{FP}} = (2^1-1)|B_2|/(2^1\cdot 2!) = 1\cdot(1/6)/(2\cdot 2) = 1/24$, using $B_2 = 1/6$; then $F_1^{\mathrm{intr}} = (c/2)(1/24) = c/48$.

For $F_2^{\mathrm{intr}}$: $\lambda_2^{\mathrm{FP}} = (2^3-1)|B_4|/(2^3\cdot 4!) = 7/(8\cdot 24)\cdot(1/30)$; since $B_4 = -1/30$, $|B_4| = 1/30$, $\lambda_2^{\mathrm{FP}} = 7\cdot(1/30)/(8\cdot 24) = 7/(5760)$. Then $F_2^{\mathrm{intr}} = (c/2)(7/5760) = 7c/11520$. □

*Remark* 4.4 (Why the $\hat{A}$-genus, not Todd). Three independent reasons force the $\hat{A}$-genus: (a) *symmetry*: the genus tower uses only even powers of $x$; the Todd class has odd powers; (b) *KO orientation*: the Hodge bundle $\mathbb{E}$ on $\overline{\mathcal{M}}_g$ carries a real structure via Serre duality ($\mathbb{E}^* \simeq \mathbb{E}\otimes\omega$); real vector bundles are oriented by KO-theory, whose multiplicative genus is $\hat{A}$; (c) *explicit integration*: the Faber–Pandharipande integrals match the $\hat{A}(ix)$ coefficients through $g = 6$.

*Remark* 4.5 (Intrinsic versus effective). The intrinsic free energies $F_g^{\mathrm{intr}} = \kappa\cdot\lambda_g^{\mathrm{FP}}$ use $\kappa(\mathrm{Vir}_c) = c/2$ and are the fundamental algebraic invariants. The *effective* free energies incorporate ghost subtraction from the $bc$ ghost system ($c_{\mathrm{ghost}} = -26$, $\kappa_{\mathrm{ghost}} = -13$):

$$F_g^{\mathrm{eff}} \;=\; \kappa_{\mathrm{eff}}\cdot\lambda_g^{\mathrm{FP}}, \qquad \kappa_{\mathrm{eff}} = \kappa + \kappa_{\mathrm{ghost}} = \frac{c-26}{2}.$$

The cancellation $\kappa_{\mathrm{eff}} = 0$ at $c = 26$ is matter-ghost cancellation (the bosonic string critical point), not vanishing of the intrinsic curvature ($\kappa(\mathrm{Vir}_{26}) = 13 \neq 0$). Throughout this paper, $F_g$ without superscript denotes $F_g^{\mathrm{intr}}$; the effective version appears only in contexts involving ghost coupling.

### 4.5. The quartic contact correction at genus 2.

**Proposition 4.6** (Genus-2 contact correction). *The quartic contact shadow* (10) *contributes a genus-*2 *correction via the $r = 4$ planted-forest graph amplitude:*

$$F_2^{\mathrm{contact}} \;=\; \mathfrak{Q}\cdot\int_{\overline{\mathcal{M}}_2}\psi_1^2 \;=\; \frac{10}{c(5c+22)}\cdot\frac{1}{240} \;=\; \frac{1}{24\,c(5c+22)},$$

*where $\int_{\overline{\mathcal{M}}_2}\psi_1^2 = 1/240$ by Faber's formula. In the semiclassical limit $c\to\infty$: $F_2^{\mathrm{contact}} \sim 1/(120\,c^2)$.*

*Remark* 4.7 (Convergence dichotomy). The shadow tower exhibits a sharp analytic dichotomy: the *scalar sector* (contact terms $\propto c/12$) has an algebraic generating function $H(t) = t^2\sqrt{Q_{\mathrm{Vir}}(t)}$ with the shadow metric $Q_{\mathrm{Vir}}(t) = c^2 + 12ct + \frac{180c+872}{5c+22}t^2$; its Taylor series converges for $|t| < R_{\mathrm{scal}} = c\sqrt{(5c+22)/(180c+872)}$. The *field sector* ($T$-dependent terms, $c$-independent) diverges factorially: $\|m_k|_T\| \sim C^k\cdot k!$ (Gevrey-1), requiring Borel resummation. The Borel sum is unambiguous because the spectral curve $u^2 = Q_{\mathrm{Vir}}(t)$ has genus 0 (no real branch points for $c > 0$): 3d gravity has no resurgent ambiguity.

## 5. Koszul duality as the algebraic shadow of S-duality

The Virasoro algebra has a Koszul dual: it is another Virasoro algebra, at the complementary central charge $26-c$. This algebraic duality is the level-2 / level-5 algebraic shadow of gravitational S-duality, computed by trace + clutching on the algebraic sector; the self-dual point $c = 13$ is the fixed point of the Koszul involution. Lifting this algebraic shadow to a nonperturbative S-duality on the dynamical-metric path integral is part of F12 (Section 12).

### 5.1. The Koszul dual.

**Theorem 5.1** (Virasoro Koszul duality). *The Koszul dual of* $\mathrm{Vir}_c$ *is* $\mathrm{Vir}_{26-c}$*:*

$$\mathrm{Vir}_c^! \;=\; \mathrm{Vir}_{26-c}.$$

*The Koszul complementarity constant is*

$$K(\mathrm{Vir}) \;=\; \kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) \;=\; \frac{c}{2} + \frac{26-c}{2} \;=\; 13. \tag{12}$$

*Self-duality holds at $c = 13$:* $\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$.

*Proof.* The bar complex $\bar{B}(\mathrm{Vir}_c)$ is a dg coalgebra. The Ran-space Verdier dual $D_{\mathrm{Ran}}(\bar{B}(\mathrm{Vir}_c))$ is a dg algebra. By the bar-cobar theorem (Theorem A of Volume I), the cobar of $D_{\mathrm{Ran}}(\bar{B})$ recovers an algebra: on the Koszul locus, this algebra is $\mathrm{Vir}_{26-c}$.

The central charge shift $c \mapsto 26 - c$ arises from the Shapovalov transposition: the Koszul dual of the Verma module at conformal weight $h$ over $\mathrm{Vir}_c$ is the contragredient Verma module at weight $h$ over $\mathrm{Vir}_{26-c}$. The shift by 26 is the $bc$ ghost central charge: the Verdier duality functor on the bar complex has the same effect as coupling to the $bc$ ghost system.

The complementarity (12) is immediate from $\kappa(\mathrm{Vir}_c) = c/2$ and $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2$. □

5.2. **The two critical central charges.** The Koszul involution $c \mapsto 26 - c$ has two distinguished points.

**Proposition 5.2** (Two critical points)**.**

(i) $c = 13$*: **Koszul self-duality.*** $\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$. *The boundary algebra equals its Koszul dual: $\kappa = \kappa' = 13/2$. The genus tower is symmetric under the involution: $F_g(c) = F_g(26-c)$ for all $g \geq 1$. The graviton resolvent $\mathcal{G}(t;\lambda)$ acquires a $c \mapsto 26 - c$ symmetry. This is the algebraic self-duality of 3d gravity; it is* not *the string theory critical point.*

(ii) $c = 26$*: **matter-ghost cancellation.** The effective curvature $\kappa_{\mathrm{eff}} = (c-26)/2 = 0$: the effective genus tower $F_g^{\mathrm{eff}} = 0$ for all $g \geq 1$ (the intrinsic tower $F_g^{\mathrm{intr}} = 13\,\lambda_g^{\mathrm{FP}} \neq 0$). The gravitational partition function becomes $S$-invariant (modular weight vanishes). This is the bosonic string critical point. The Koszul dual* $\mathrm{Vir}_{26}^! = \mathrm{Vir}_0$ *is the Witt algebra at $c = 0$: uncurved ($\kappa = 0$) but non-formal (infinite $A_\infty$ tower from the field-dependent terms).*

*Proof.* Part (i): at $c = 13$, the involution $c \mapsto 26 - c$ fixes $c$, so $\mathrm{Vir}_{13}^! = \mathrm{Vir}_{26-13} = \mathrm{Vir}_{13}$. The symmetry $F_g(c) = F_g(26-c)$ follows from $F_g = (c/2)\lambda_g^{\mathrm{FP}}$: replacing $c$ by $26 - c$ gives $((26-c)/2)\lambda_g^{\mathrm{FP}}$; the two are equal when $c/2 = (26-c)/2$, i.e. $c = 13$.

Part (ii): at $c = 26$, $\kappa_{\mathrm{eff}} = (c-26)/2 = 0$ directly, so $F_g^{\mathrm{eff}} = 0$. The effective genus-1 partition function $Z_1^{\mathrm{eff}}(\tau) = \eta(\tau)^{-\kappa_{\mathrm{eff}}} = \eta(\tau)^0 = 1$. The Koszul dual is $\mathrm{Vir}_0$; by Theorem 2.10(iii), $\mathrm{Vir}_0$ has $\kappa = 0$ (uncurved) but $m_k \neq 0$ for all $k \geq 3$ (non-formal). □

5.3. **The complementarity constant and Page time.** The complementarity constant $K = 13$ has physical content beyond its algebraic origin.

**Proposition 5.3** (Complementarity and the genus tower)**.** *For any $g \geq 1$:*

$$F_g(\mathrm{Vir}_c) + F_g(\mathrm{Vir}_{26-c}) \;=\; 13 \cdot \lambda_g^{\mathrm{FP}}.$$

*The sum is independent of $c$: the total intrinsic genus-$g$ free energy of the Koszul pair is a topological constant equal to the complementarity constant $K = 13$ times the Hodge integral.*

*Proof.* $F_g(\mathrm{Vir}_c) + F_g(\mathrm{Vir}_{26-c}) = \frac{c}{2}\lambda_g^{\mathrm{FP}} + \frac{26-c}{2}\lambda_g^{\mathrm{FP}} = 13\,\lambda_g^{\mathrm{FP}}$. □

*Remark* 5.4 (Verdier duality and the modular $S$-transform). The Koszul involution $\mathbb{D}\colon c \mapsto 26 - c$ (Verdier duality on the bar complex) and the modular $S$-transform $S\colon \tau \mapsto -1/\tau$ commute on genus-1 MC elements: $S \circ \mathbb{D}(\Theta^{(1)}(\tau)) = \mathbb{D} \circ S(\Theta^{(1)}(\tau)) = \Theta^{(1)}_{\mathrm{Vir}_{26-c}}(-1/\tau)$. They act on different variables ($c$ vs. $\tau$). The commutativity produces a paired Cardy formula: $S_{\mathrm{Cardy}}(\mathrm{Vir}_c; h) + S_{\mathrm{Cardy}}(\mathrm{Vir}_{26-c}; h) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{(26-c)h/6}$, which is maximised at $c = 13$ by concavity of $\sqrt{\cdot}$.

### 5.4. The quartic contact under Koszul duality.

**Proposition 5.5** (Quartic complementarity). *Under the Koszul involution $c \mapsto 26 - c$:*

$$\mathfrak{Q}(c) + \mathfrak{Q}(26 - c) \;=\; \frac{10}{c(5c + 22)} + \frac{10}{(26 - c)(152 - 5c)}.$$

*At $c = 13$ (self-dual): $\mathfrak{Q}(13) = 10/(13 \cdot 87) = 10/1131$. The quartic contact invariant is the first nonlinear witness of the duality between boundary and line sectors.*

*Remark* 5.6 (The gauge-gravity-matter trichotomy). The complexity classification by the discriminant $\Delta$ organises the standard landscape into three columns:

| | **Gauge (CS/KM)** | **Pure gravity (Vir/$\mathcal{W}_N$)** | **Matter-coupled (BP)** |
|---|---|---|---|
| Class | **L** | **M** | **M** |
| Shadow depth | 3 | $\infty$ | $\infty$ |
| $A_\infty$ tower | finite | infinite | infinite |
| $\Delta$ | 0 | $\neq 0$ | $\neq 0$ |

The passage from gauge ($\Delta = 0$, integrable) to gravity ($\Delta \neq 0$, chaotic) is Drinfeld–Sokolov reduction: the Sugawara construction manufactures the quartic pole from double poles, and **L** $\to$ **M**.

## 6. The holographic datum $\mathcal{H}(\mathrm{Vir}_c)$

The six preceding sections assembled the gravitational Koszul data from the single input (2). We now organise the output into the *holographic modular Koszul datum*: a seven-entry algebraic package. It becomes a physical bulk datum only after the open–closed comparison and the line-comparison maps are supplied.

### 6.1. The seven-entry package.

**Definition 6.1** (Holographic datum of $\mathrm{Vir}_c$). The *holographic datum* of 3d gravity at central charge $c = 3\ell/(2G)$ is the seven-entry package

$$\mathcal{H}(\mathrm{Vir}_c) \;=\; \big(\mathrm{Vir}_c,\ \mathrm{Vir}_c^{\mathrm{i}},\ \mathrm{Vir}_{26-c},\ Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c),\ r_c(z),\ \Theta_c,\ \nabla^{\mathrm{hol}}\big),$$

where the seven components are:

- (i) $\mathrm{Vir}_c$: the **boundary algebra**, the Virasoro vertex algebra at central charge $c = 3\ell/(2G)$, with OPE (3) and $\kappa(\mathrm{Vir}_c) = c/2$;
- (ii) $\mathrm{Vir}_c^{\mathrm{i}} = H^*(\bar{B}(\mathrm{Vir}_c))$: the **bar-cohomology coalgebra**, before Verdier or linear duality;
- (iii) $\mathrm{Vir}_{26-c}$: the **Koszul dual**, the line-operator algebra (Theorem 5.1), with $\kappa(\mathrm{Vir}_{26-c}) = (26 - c)/2$;
- (iv) $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$: the **algebraic closed-sector object**, not the physical bulk until the open–closed comparison is proved. The line category is $\mathrm{Vir}_{26-c}$-$\mathsf{mod}$ on the strict line-comparison surface; it is attached to the third entry.
- (v) $r_c(z) = (c/2)/z^3 + 2T/z$: the **classical $r$-matrix** (Proposition 2.5), the collision residue of the Virasoro $\lambda$-bracket with cubic-plus-simple poles;
- (vi) $\Theta_c$: the **universal Maurer–Cartan element** in the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathrm{Vir}_c}$, whose projections constitute the shadow obstruction tower $\{S_r\}_{r\geq 2}$ (Definition 4.1);
- (vii) $\nabla^{\mathrm{hol}}$: the **shadow connection**, the flat connection on the configuration-space bundle determined by the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

*Remark* 6.2 (The algebraic closed/open/line triangle). Entries (i), (iii), and (iv) form the *algebraic closed/open/line triangle*:

$$\begin{array}{rcl} \textbf{Closed sector} &=& Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) \simeq \mathbb{C}[[c]] \\ \textbf{Boundary} &=& \mathrm{Vir}_c \\ \textbf{Lines} &=& \mathrm{Vir}_{26-c}\text{-mod (on Koszul locus)} \end{array}$$

The arrows are: bulk $\xrightarrow{\beta_B}$ boundary (the conformal Ward identity (9)), boundary $\xrightarrow{\bar{B}}$ lines (the bar complex followed by Verdier duality gives $\mathrm{Vir}_{26-c}$-modules), lines $\xrightarrow{Z^{\mathrm{der}}}$ bulk (chiral Hochschild cochains close the triangle).

6.2. **Explicit computation of the seven-entry package.** The seven-entry package $\mathcal{H}(\mathrm{Vir}_c)$ is determined by $c$. Its nonformal numerical projections admit closed-form expressions.

**Theorem 6.3** (The holographic datum: explicit form)**.** *The seven entries of $\mathcal{H}(\mathrm{Vir}_c)$ are determined by the single datum c; their numerical projections include:*

- (i) *Boundary curvature: $\kappa(\mathrm{Vir}_c) = c/2$; dual curvature: $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2$; complementarity: $\kappa + \kappa' = 13$.*
- (ii) *Shadow obstruction tower through degree 5:*

$$\begin{aligned} S_2 &= \kappa = c/2 && \textit{(curvature)},\\ S_3 &= -c && \textit{(cubic shadow)},\\ S_4 &= \frac{10}{c(5c+22)} && \textit{(quartic contact)},\\ S_5 &= \frac{-48}{c^2(5c+22)} && \textit{(quintic)}. \end{aligned}$$

  *At large c (semiclassical): $S_4 \sim 2/c^2 \sim G^2$, $S_5 \sim -48/(5c^3) \sim G^3$.*
- (iii) *Classical r-matrix: $r_c(z) = (c/2)/z^3 + 2T/z$; at $c = 0$: $r_0(z) = 2T/z$ (field-dependent only, no central extension); at $c = 13$: $r_{13}(z) = (13/2)/z^3 + 2T/z$ (self-dual).*
- (iv) *Spectral coproduct: strictly primitive, $\Delta_z(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. No higher terms at any degree: gravitons do not split.*
- (v) *Shadow connection: $\nabla^{\mathrm{hol}} = d - \sum_{r\geq 2} S_r\, \omega_{(r)}$, where $\omega_{(r)}$ is the degree-r component of the bar-complex propagator form on the configuration space $\mathrm{FM}_n(\mathbb{C})$. Flatness of $\nabla^{\mathrm{hol}}$ is equivalent to the MC equation for $\Theta_c$.*
- (vi) *Discriminant: $\Delta = 8\kappa S_4 = 40/(5c+22) \neq 0$ for $c \neq -22/5$. Class **M**: infinite shadow depth.*

*Proof.* Each listed projection was computed in the preceding sections: (i) equation (12); (ii) Definition 4.1 for $S_2$–$S_4$, and the quintic shadow from the shadow-metric expansion $H(t) = t^2\sqrt{Q_{\mathrm{Vir}}(t)}$ with $Q_{\mathrm{Vir}}(t) = c^2 + 12ct + (180c + 872)t^2/(5c+22)$, giving $S_5 = [t^5]H/5 = -48/[c^2(5c+22)]$; (iii) equation (8); (iv) the DS ghost-number obstruction (Section 2); (v) the MC equation of Section 4; (vi) equation (11). □

6.3. **The shadow connection and soft graviton hierarchy.** The shadow connection $\nabla^{\mathrm{hol}}$ decomposes the MC element into a hierarchy of Ward identities. The first three are classical; the rest are new.

**Proposition 6.4** (Shadow connection and soft gravitons)**.** *The shadow connection $\nabla^{\mathrm{hol}}$ on $\mathrm{FM}_n(\mathbb{C})$ acts on n-point correlation functions of primary operators $O_i$ of weight $h_i$. Its decomposition by shadow degree gives:*

- (i) *Degree 2 ($S_2 = c/2$): the Weinberg leading soft graviton theorem. Soft factor: $S_i^{(0)} = \partial_{z_i}/(z_0 - z_i)$.*

(ii) *Degree 2 (double pole of $S_2$): the Cachazo–Strominger subleading soft theorem. Soft factor:* $S_i^{(1)} = h_i/(z_0 - z_i)^2$.

(iii) *Degree 3–4: the cubic shadow* $S_3 = -c$ *is gauge-trivial; the first non-trivial sub-subleading contribution is the quartic contact* $S_4 = 10/[c(5c + 22)]$, *the Cachazo–He–Yuan soft factor requiring the ternary* $A_\infty$ *operation* $m_3$.

(iv) *Degree* $r \geq 5$*: one independent Ward identity per degree, forced by the class* **M** *property* ($m_k \neq 0$ *for all* $k \geq 3$). *The infinite* $A_\infty$ *tower produces infinitely many soft theorems beyond the known three.*

*At large c: the soft factors at degree r scale as* $S_r \sim G^{r-2}$, *and the semiclassical limit retains only* $S_2$ *(Weinberg) and the subleading* $S^{(1)}$ *(Cachazo–Strominger).*

*Proof.* The shadow connection is flat (from the MC equation), so each degree-$r$ component $S_r$ gives an independent Ward identity. The degree-2 components extract the conformal Ward identity, whose simple-pole and double-pole residues are the Weinberg and Cachazo–Strominger soft factors. The cubic shadow is gauge-trivial: $S_3 = d_2\sigma_3$ for an explicit gauge transformation $\sigma_3$ (using the fact that the Virasoro algebra is simple for generic $c$, so $H^1(\mathrm{Vir}_c, \mathrm{Vir}_c) = 0$). The quartic contact $S_4 = 10/[c(5c + 22)]$ is genuinely new by the non-vanishing of the Kac determinant at level 4. For degree $\geq 5$: non-vanishing follows from the infinite tower (Theorem 2.10). □

## 6.4. The spectral braiding.

**Proposition 6.5** (Virasoro fusion kernel as braiding). *The braided monoidal structure on* $\mathbf{C}_c$ *is determined by the r-matrix* $r_c(z) = (c/2)/z^3 + 2T/z$ *via the KZ-type connection on the category of Virasoro modules:*

$$\nabla_{\mathrm{KZ}} \;=\; d - \sum_{i<j} r_c(z_i - z_j)\, dz_{ij}, \tag{13}$$

*whose monodromy is the Virasoro fusion kernel (the Ponsot–Teschner* $6j$*-symbol* [5]*). The KZ connection has an irregular singularity at* $z_i = z_j$ *of Poincaré rank 2 (from the cubic pole), in contrast to the regular-singular KZ connection of affine Kac–Moody algebras (where the r-matrix has only a simple pole). The irregular singularity produces Stokes phenomena in the braiding, not present in the KM setting.*

*Proof.* The classical $r$-matrix (8) determines the connection 1-form $\sum r_c(z_{ij})\, dz_{ij}$ on $\mathrm{Conf}_n(\mathbb{C})$. The CYBE (Proposition 2.7) is equivalent to flatness of this connection. The Poincaré rank at $z_i = z_j$ is the order of the pole minus one: $(z_{ij})^{-3}$ gives rank 2. The monodromy of the flat connection around $z_j = z_i + \epsilon\, e^{2\pi i\theta}$ produces the braiding on modules; for the Virasoro algebra, this is the Ponsot–Teschner kernel. □

*Remark* 6.6 (Gravity versus gauge theory). The holographic datum distinguishes gravity from gauge theory through three structural signatures:

| | **Gauge** ($V_k(\mathfrak{g})$) | **Gravity** ($\mathrm{Vir}_c$) |
|---|---|---|
| $r$-matrix pole | simple ($1/z$) | cubic ($1/z^3$) |
| Coproduct | non-trivial | primitive |
| $A_\infty$ tower | finite ($m_k = 0$, $k \geq 3$) | infinite (all $m_k \neq 0$) |
| KZ singularity | regular | irregular (Poincaré rank 2) |
| Shadow depth | 3 (class **L**) | $\infty$ (class **M**) |
| Discriminant | $\Delta = 0$ | $\Delta \neq 0$ |

The passage from gauge to gravity is Drinfeld–Sokolov reduction: the Sugawara construction manufactures the quartic pole from double poles, simultaneously creating the infinite $A_\infty$ tower, the irregular KZ singularity, and the primitive coproduct.

## 7. The genus expansion and the $\hat{A}$-genus

The shadow obstruction tower of Section 4 produces a sequence of genus-$g$ free energies $F_g$ through the scalar genus tower theorem. This section derives the same free energies from an independent method: the Grothendieck–Riemann–Roch theorem applied to the Hodge bundle on $\overline{\mathcal{M}}_g$. The agreement of the two derivations is a non-trivial consistency check.

### 7.1. The genus-1 formal coefficient and the Weinberg soft theorem.

**Theorem 7.1** (Genus-1 formal coefficient). *The genus-1 formal scalar coefficient of* $\mathrm{Vir}_c$ *is (UNIFORM-WEIGHT):*

$$F_1(\mathrm{Vir}_c) \;=\; \frac{\kappa(\mathrm{Vir}_c)}{24} \;=\; \frac{c}{48}.$$

*Three independent derivations:*

(i) ***Shadow tower.*** $F_1 = \kappa \cdot \lambda_1^{\mathrm{FP}}$*, where* $\lambda_1^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_{1,1}} \lambda_1 = 1/24$ *and* $\kappa(\mathrm{Vir}_c) = c/2$.

(ii) ***Dedekind eta function.*** *The genus-1 partition function is* $Z_1(\tau) = \mathrm{Tr}_{\mathrm{Vir}_c}(q^{L_0 - c/24}) = q^{-c/24}/\prod_{n\geq 1}(1-q^n) = q^{-c/24}\,\eta(\tau)^{-1}$*, where* $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ *is the Dedekind eta function. The free energy* $F_1 = -\log|Z_1|^2|_{q\to 0} = c/48 \cdot \log|q|^{-2} = c/48$ *(leading term).*

(iii) ***Soft graviton.*** *The Weinberg leading soft factor* $S_i^{(0)} = \partial_{z_i}/(z_0 - z_i)$ *applied to the genus-1 one-point function gives* $\langle T\rangle_{g=1} = -c/24$ *(the Casimir energy on the torus). The free energy* $F_1 = \kappa \cdot \lambda_1^{\mathrm{FP}} = (c/2)(1/24) = c/48$.

*Proof.* Derivation (i) follows from the genus tower (Theorem 4.3) at $g = 1$, using the intrinsic curvature $\kappa = c/2$ (not the effective curvature).

Derivation (ii): the vacuum character of $\mathrm{Vir}_c$ for generic $c$ (no null vectors) is $\chi_0(\tau) = q^{-c/24}\prod_{n\geq 2}(1-q^n)^{-1}$. The full partition function on a genus-1 surface with modular parameter $\tau$ includes the full Fock space: $Z_1(\tau) = q^{-c/24}\prod_{n\geq 1}(1-q^n)^{-1}$. The leading $q$-expansion gives the free energy $F_1 = c/48$.

Derivation (iii): the genus-1 one-point function is

$$\langle T\rangle_{g=1} \;=\; \frac{\mathrm{Tr}(L_0\, q^{L_0 - c/24})}{\mathrm{Tr}(q^{L_0 - c/24})} \;=\; q\,\partial_q \log Z_1(\tau) \;=\; -\frac{c}{24} + O(q).$$

This is the Casimir energy density. The integrated free energy is $F_1 = (c/2)\cdot(1/24)$. □

### 7.2. The genus-2 formal coefficient.

**Theorem 7.2** (Genus-2 formal coefficient). *The genus-2 formal scalar coefficient is (UNIFORM-WEIGHT):*

$$F_2(\mathrm{Vir}_c) \;=\; \frac{7\,\kappa(\mathrm{Vir}_c)}{5760} \;=\; \frac{7c}{11520}.$$

*The Hodge integral value* $\lambda_2^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_2} \lambda_2 = 7/5760$ *uses* $B_4 = -1/30$ *and* $(2^3-1)/2^3 = 7/8$.

*Proof.* From the genus tower (Theorem 4.3): $F_2 = \kappa\cdot\lambda_2^{\mathrm{FP}} = (c/2)(7/5760) = 7c/11520$. The computation of $\lambda_2^{\mathrm{FP}}$: $\lambda_2^{\mathrm{FP}} = (2^{2\cdot 2-1}-1)|B_{2\cdot 2}|/(2^{2\cdot 2-1}\cdot(2\cdot 2)!) = (2^3-1)|B_4|/(2^3\cdot 4!) = 7\cdot(1/30)/(8\cdot 24) = 7/5760$. □

*Remark* 7.3 (The quartic contact correction). The shadow free energy $F_2 = 7\kappa/5760$ is the scalar-lane contribution. The quartic contact shadow adds a correction (Proposition 4.6): $F_2^{\mathrm{contact}} = \mathfrak{Q}\cdot\int_{\overline{\mathcal{M}}_2}\psi_1^2 = 10/[c(5c+22)]\cdot 1/240 = 1/[24c(5c+22)]$. At large $c$: $F_2^{\mathrm{contact}} \sim 1/(120c^2)$, which is $O(G^2)$, a genuine two-loop quantum gravity correction invisible in the one-loop $\hat{A}$-genus formula.

7.3. **The GRR derivation.** The $\hat{A}$-genus formula for the genus tower admits an independent derivation from the Grothendieck–Riemann–Roch theorem, providing a second proof that does not depend on the shadow obstruction tower.

**Theorem 7.4** (GRR derivation of the genus tower). *Let $\pi : \mathcal{C}_g \to \overline{\mathcal{M}}_g$ be the universal curve, and $\mathbb{E} = \pi_* \omega_{\mathcal{C}_g/\overline{\mathcal{M}}_g}$ the Hodge bundle. The Grothendieck–Riemann–Roch theorem gives:*

$$\mathrm{ch}(\mathbb{E}) \;=\; \pi_*\big(\mathrm{td}(\omega_{\mathcal{C}_g/\overline{\mathcal{M}}_g})\big) \;=\; g + \sum_{k\geq 1} \frac{B_{2k}}{(2k)!}\,\kappa_{2k-1},$$

*where $\kappa_{2k-1}$ is the Miller–Morita–Mumford class and $B_{2k}$ are Bernoulli numbers. The generating function of the Hodge integrals $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g$ is determined by the $\hat{A}$-genus:*

$$\sum_{g\geq 1} \lambda_g^{\mathrm{FP}}\, x^{2g} \;=\; \hat{A}(ix) - 1, \qquad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}.$$

*Proof.* The Chern character of the Hodge bundle is computed by GRR: $\mathrm{ch}(\mathbb{E}) = \pi_*(\mathrm{td}(\Omega^1_\pi))$, where td is the Todd class. The top Chern class $\lambda_g = c_g(\mathbb{E})$ is extracted from $\mathrm{ch}(\mathbb{E})$ via the Newton identities relating Chern characters to Chern classes. The generating function of $\lambda_g^{\mathrm{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g$ is then determined by integrating the Todd class over the fibres of $\pi$: the fibre integral produces the $\hat{A}$-genus of the tangent bundle of $\mathcal{C}_g$ (which has real rank 2, so the KO orientation produces $\hat{A}$ rather than Todd). □

*Remark* 7.5 (Why $\hat{A}$, not Todd: three reasons). The GRR derivation confirms Remark 4.4:

(a) *Even powers*: the genus tower uses $x^{2g}$ (even); the Todd class has odd powers.
(b) *KO orientation*: the Hodge bundle carries a real structure from Serre duality $\mathbb{E}^* \simeq \mathbb{E} \otimes \omega$; the real vector bundle is oriented by KO-theory, whose multiplicative genus is $\hat{A}$.
(c) *Explicit verification*: the Faber–Pandharipande integrals match $\hat{A}(ix)$ through $g = 6$.

7.4. **Uniform-weight structure for the Virasoro algebra.**

**Proposition 7.6** (Uniform-weight and absence of cross-channel corrections). *The Virasoro algebra* $\mathrm{Vir}_c$ *has a single generating field $T(z)$ of conformal weight 2. The genus-$g$ free energy $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) receives no cross-channel correction: $\delta F_g^{\mathrm{cross}} = 0$ for all $g \geq 1$.*

*Proof.* The cross-channel correction $\delta F_g^{\mathrm{cross}}$ arises from the interaction between fields of different conformal weights in the propagator sum. The Virasoro algebra has a single generator of weight 2: there is only one channel, and the cross-channel correction vanishes identically. The scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is exact at all genera for this single-weight algebra. □

7.5. **The genus-1 fibre differential.** The explicit genus-1 bar complex provides a direct computation of $F_1$. The genus-1 propagator on the elliptic curve $E_\tau$ is $\eta^{(1)}(z_1, z_2;\, \tau) \;=\; \partial_{z_1} \log \vartheta_1(z_1 - z_2 \,|\, \tau) + 2\pi i\, \mathrm{Im}(z_1 - z_2)/\mathrm{Im}\,\tau$.

**Proposition 7.7** (The three-term formula). *The fibre differential on the degree-2 bar element $[s^{-1}T \,|\, s^{-1}T]$ at genus 1 is:*

$$d_{\mathrm{fib}}^{(1)}[s^{-1}T \,|\, s^{-1}T] \;=\; s^{-1}\!\left(:TT: \;-\; \frac{2\pi^2}{3}\, E_2(\tau)\cdot T \;-\; \frac{c\pi^4}{90}\, E_4(\tau)\cdot \mathbf{1}\right),$$

*where $E_2(\tau) = 1 - 24\sum_{n\geq 1} \sigma_1(n)\, q^n$ is the quasi-modular Eisenstein series and $E_4(\tau)$ is the weight-4 Eisenstein series. The three terms arise from:*

(i) $:TT:$: *the genus-*0 *normal-ordered product (regular term paired with* $1/u$*);*
(ii) $-\frac{2\pi^2}{3}E_2(\tau)\cdot T$*: the genus-*1 *state-dependent correction (double pole* $2T/u^2$ *paired with the propagator coefficient* $-(\pi^2/3)E_2(\tau)\,u$*);*
(iii) $-\frac{c\pi^4}{90}E_4(\tau)\cdot\mathbf{1}$*: the genus-*1 *vacuum correction (quartic pole* $(c/2)/u^4$ *paired with* $-(\pi^4/45)E_4(\tau)\,u^3$*).*

*Proof.* The meromorphic part of the genus-1 propagator has the Laurent expansion $h(u;\,\tau) = 1/u - (\pi^2/3)E_2(\tau)\,u - (\pi^4/45)E_4(\tau)\,u^3 - \cdots$ (odd powers only). The fibre differential is the residue of $h(u;\,\tau)\cdot T(z_1)\,T(z_2)$ at $u = 0$. The Virasoro OPE (3) has poles at orders 4, 2, and 1 in $u = z_1 - z_2$. Pairing the propagator terms $c_k(\tau)\,u^{2k-1}$ with the OPE terms $T_{(2k-1)}T\,u^{-(2k)}$ gives the residue: $T_{(-1)}T = :TT:$ from the leading $1/u$, $c_1\cdot T_{(1)}T = -(\pi^2/3)E_2\cdot 2T$ from $k=1$, $c_2\cdot T_{(3)}T = -(\pi^4/45)E_4\cdot(c/2)$ from $k=2$. □

The $E_2$-correction is quasi-modular (it transforms anomalously under $\tau\mapsto -1/\tau$); the $E_4$-correction is fully modular. The Virasoro OPE pole structure generates precisely the Eisenstein tower $\{E_{2k}\}$ at genus 1.

## 8. BTZ black holes as Maurer–Cartan elements

The BTZ black hole of Bañados, Teitelboim, and Zanelli [4] is the unique black hole solution of 2+1-dimensional gravity with $\Lambda < 0$. We identify it as a Maurer–Cartan element of the gravitational convolution algebra of the algebraic sector. The identification is unconditional at the algebraic level (boundary $A_b = \mathrm{Vir}_c$, MC element $\alpha_{\mathrm{BTZ}}$); the lift of $\alpha_{\mathrm{BTZ}}$ to a saddle of a compact dynamical-metric path integral, and the recovery of microstate counts from a sum over geometries, requires the BTZ / Cardy hypothesis package and the Maloney–Witten + Manschot–Moore Farey-tail completion (Section 12).

**BTZ / Cardy hypothesis package** (named for use throughout this section and Section 9):

(BC1) *Modular invariance.* The torus partition function $Z_1(\tau)$ is invariant under $\mathrm{SL}_2(\mathbb{Z})$, and in particular under the $S$-transform $\tau\mapsto -1/\tau$ used in the Cardy saddle-point.

(BC2) *Vacuum dominance.* As $\tau\to i0^+$ along the imaginary axis, $Z_1(\tau)$ is dominated by the Virasoro vacuum representation: $Z_1(\tau)\sim e^{-2\pi i(-c/24)\tau}\cdot(1+O(q))$.

(BC3) *Saddle stability.* The saddle-point approximation to the modular $S$-transform is uncorrected at leading order in $h \gg c$, with subleading corrections enumerated by the Hardy–Ramanujan circle method (Faulkner–Lewkowycz subleading; Manschot–Moore Farey expansion).

(BC4) *Maloney–Witten Farey-tail completion.* The sum over $\mathrm{SL}_2(\mathbb{Z})/\mathbb{Z}$ images of the BTZ saddle produces the Maloney–Witten partition function $Z_{\mathrm{MW}}(\tau)$; the Manschot–Moore Farey-tail prescription regularises this (otherwise non-convergent / negatively-graded) sum into a holomorphic mock-modular completion.

The identification of $\alpha_{\mathrm{BTZ}}$ as an MC element is unconditional. The Cardy entropy formula (Theorem 8.3) below uses (BC1)–(BC3); the genus-$g$ saddle expansion (Proposition 8.4) uses (BC3); the construction of a compact gravitational partition function from the algebraic sector uses all four hypotheses and is the F12 open problem.

### 8.1. The BTZ solution.

A BTZ black hole is specified by two parameters: mass $M$ and angular momentum $J$, subject to $|J|\le M\ell$. In the Chern–Simons formulation, the BTZ solution is a pair of flat $\mathfrak{sl}_2$-connections with holonomy around the spatial circle:

**Definition 8.1** (BTZ gauge configuration)**.** The BTZ black hole with mass $M$ and angular momentum $J$ is the flat connection $A_\pm$ with holonomy eigenvalues

$$\exp\big(\pm 2\pi\sqrt{h}\,\big) \qquad \text{and} \qquad \exp\big(\pm 2\pi\sqrt{\bar h}\,\big),$$

where the conformal weights are

$$h = \frac{M\ell + J}{2}, \qquad \bar h = \frac{M\ell - J}{2}.$$

The outer horizon radius is $r_+ = \ell\sqrt{8GM}\big(1 + \sqrt{1 - (J/(M\ell))^2}\big)^{1/2}$.

### 8.2. BTZ as MC element.

**Theorem 8.2** (BTZ as Maurer–Cartan element)**.** ***Type signature.*** *Quadrant: Open. Presentation: chain-level (Maurer–Cartan in the modular convolution algebra of the algebraic sector). Level: 4 (operator-level: line / flat-connection encoding). Hypothesis package:* $\Lambda < 0$ *($AdS_3$); flat* $A_\pm$ *connection on the solid torus* $D \times S^1$ *with prescribed holonomy; identification of holonomy weights with conformal weights* $(h, \bar h)$ *via Brown–Henneaux. The lift to a saddle of a compact dynamical-metric path integral requires (BC1)–(BC4) and is the F12 frontier.*

*The BTZ black hole of mass* $M$ *and angular momentum* $J$ *is a Maurer–Cartan element*

$$\alpha_{\mathrm{BTZ}} = h\, e_h + \bar h\, e_{\bar h} \in \mathfrak{g}^{\mathrm{mod}}_{\mathrm{Vir}_c},$$

*where* $e_h, e_{\bar h}$ *are the generators of the modular convolution algebra at conformal weights* $h = (M\ell + J)/2$, $\bar h = (M\ell - J)/2$*. The MC equation* $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ *is equivalent to the flatness condition* $F_\pm = 0$ *for the BTZ gauge connection.*

*Proof.* The BTZ solution is a flat $SL(2, \mathbb{R})$ connection on the solid torus $D \times S^1$ with specified holonomy. In the boundary Virasoro description, flatness becomes the MC equation: the highest-weight state $|h, \bar h\rangle$ in the Virasoro module of weight $(h, \bar h)$ satisfies the conformal Ward identities, which are exactly the components of the MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ projected to the boundary. □

### 8.3. The Cardy entropy.

The statistical entropy of the BTZ black hole is computed from the asymptotic growth of Virasoro characters.

**Theorem 8.3** (Cardy formula and Bekenstein–Hawking entropy)**.** ***Type signature.*** *Quadrant: Open. Presentation: chain-level + Cardy saddle. Level: 5 (modular trace of level-4 operators). Hypothesis package: BTZ / Cardy (BC1)–(BC3) above; trace direction (operators* $\to$ *scalar) only.*

*The microstate entropy of the BTZ black hole at conformal weights* $(h, \bar h)$ *with* $h, \bar h \gg c$ *is:*

$$S_{\mathrm{Cardy}}(h, \bar h) = 2\pi\sqrt{\frac{c\,h}{6}} + 2\pi\sqrt{\frac{c\,\bar h}{6}} = S_{\mathrm{BH}},$$

*where the Bekenstein–Hawking entropy is*

$$S_{\mathrm{BH}} = \frac{\pi r_+}{2G} = \frac{\mathrm{Area(horizon)}}{4G}.$$

*Proof.* The Cardy formula: the number of Virasoro primaries of weight $h$ grows as $\rho(h) \sim \exp(2\pi\sqrt{ch/6})$ for $h \gg c$ (the saddle-point approximation to the modular $S$-transform of the partition function). For the BTZ state $(h, \bar h)$: $S = \log\rho(h) + \log\rho(\bar h) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{c\bar h/6}$.

The Bekenstein–Hawking entropy in terms of boundary data: using $c = 3\ell/(2G)$, $h = (M\ell + J)/2$, $\bar{h} = (M\ell - J)/2$:

$$\begin{aligned} S_{\text{Cardy}} &= 2\pi\sqrt{\frac{3\ell}{12G}(M\ell + J)} + 2\pi\sqrt{\frac{3\ell}{12G}(M\ell - J)} \\ &= 2\pi\sqrt{\frac{\ell}{4G} \cdot \frac{M\ell + J}{1}} + 2\pi\sqrt{\frac{\ell}{4G} \cdot \frac{M\ell - J}{1}} \\ &= \frac{\pi r_+}{2G}, \end{aligned}$$

where the last step uses the relation $r_\pm^2 = 8G\ell^2 M\big(1 \pm \sqrt{1 - (J/(M\ell))^2}\big)/2$ for the non-rotating case $J = 0$: $r_+ = \ell\sqrt{8GM}$, $S_{\text{BH}} = \pi\ell\sqrt{8GM}/(2G) = \pi r_+/(2G)$. □

8.4. **Quantum corrections to BTZ entropy.** The shadow obstruction tower provides a systematic genus expansion of quantum corrections to the leading Bekenstein–Hawking entropy.

**Proposition 8.4** (BTZ genus expansion)**.** *The entropy of the BTZ black hole admits a saddle-point expansion:*

$$S(M) \;=\; S_{\text{BH}} - \frac{3}{2}\log S_{\text{BH}} + \sum_{g=2}^{\infty} \frac{C_g(\kappa)}{S_{\text{BH}}^{2g-2}},$$

*where* $S_{\text{BH}} = 2\pi\sqrt{ch/6}$ *and the coefficients* $C_g(\kappa)$ *depend on the shadow free energies* $F_1, \ldots, F_g$*:*

(i) *The leading correction is logarithmic:* $-\frac{3}{2}\log S_{\text{BH}}$*, universal across all* 2+1*-dimensional black holes.*
(ii) $F_1 = \kappa/24 = c/48$*: the one-loop free energy from the scalar shadow.*
(iii) $F_2 = 7\kappa/5760 = 7c/11520$*: the two-loop free energy.*
(iv) *The shadow partition function* $Z^{\text{sh}} = \exp(\sum_{g\geq 1} F_g \quad {}^{2g})$ *converges absolutely (Bernoulli decay* $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$*), in contrast to the string free energy which diverges as* $(2g)!$*.*

*Proof.* The saddle-point expansion of $Z(\beta) = \exp(\sum_{g\geq 0} F_g\,\beta^{2g})$ around the BTZ saddle gives the $1/S_{\text{BH}}$ expansion. The logarithmic correction $-3/2$ is the universal contribution from the one-loop determinant of the graviton fluctuation around the BTZ geometry (the Jacobian of the saddle-point integral in three dimensions). The higher-order corrections $C_g$ are determined by the shadow free energies through the saddle-point formula. □

8.5. **BTZ complementarity and the Hawking–Page transition.**

**Proposition 8.5** (BTZ paired entropy)**.** *The Koszul pair* $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$ *induces a paired entropy:*

$$S_{\text{BH}}(\mathrm{Vir}_c;\, h) + S_{\text{BH}}(\mathrm{Vir}_{26-c};\, h) \;=\; 2\pi\sqrt{\frac{h}{6}}\Big(\sqrt{c} + \sqrt{26-c}\Big),$$

*maximised at* $c = 13$ *by the concavity of* $\sqrt{\cdot}$*, where* $\sqrt{c} + \sqrt{26-c}\,\big|_{c=13} = 2\sqrt{13}$*. The Hawking–Page transition at* $\beta_{\text{HP}} = 2\pi\ell$ *exchanges the BTZ and thermal* AdS *saddles.*

*Proof.* The paired entropy is $S_{\text{BH}}(\mathrm{Vir}_c) + S_{\text{BH}}(\mathrm{Vir}_{26-c}) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{(26-c)h/6} = 2\pi\sqrt{h/6}(\sqrt{c} + \sqrt{26-c})$. By AM-QM inequality, $\sqrt{c} + \sqrt{26-c}$ is maximised when $c = 26 - c$, i.e. $c = 13$. The Hawking–Page temperature $\beta_{\text{HP}} = 2\pi\ell$ is the point where the BTZ free energy $F_{\text{BTZ}} = -(c/6\pi)(\pi^2/\beta)$ crosses the thermal AdS free energy $F_{\text{AdS}} = -(c/6\pi)\beta$. □

*Remark* 8.6 (The entanglement entropy). The Calabrese–Cardy entanglement entropy of a boundary interval of length $L$ is $S_{\mathrm{EE}} = (c/3)\log(L/\varepsilon)$. The Koszul-dual entanglement entropy is $S'_{\mathrm{EE}} = ((26-c)/3)\log(L/\varepsilon)$. Their sum: $S_{\mathrm{EE}} + S'_{\mathrm{EE}} = (26/3)\log(L/\varepsilon)$, independent of $c$. At $c = 13$: $S_{\mathrm{EE}} = S'_{\mathrm{EE}} = (13/3)\log(L/\varepsilon)$.

## 9. The algebraic Page-curve shadow from complementarity

Under the BTZ / Cardy hypothesis package (Section 8, (BC1)–(BC3)) and the island-saddle mechanism upstream (Penington [6], Almheiri–Engelhardt–Marolf–Maxfield [7, 8]), the complementarity constant $K = \kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$ controls the transition between the two Koszul-dual phases of the algebraic-sector saddle competition. The derivation below is the algebraic Page shadow of the gravitational Page curve: the Lagrangian decomposition of the genus expansion evaluated on the BTZ Maurer–Cartan element, the evaporation rate from the genus-1 free energy, and the Page time as the crossing of the two saddle-point contributions on the algebraic sector. The $c$-independent Page time $t_P = 3S_{\mathrm{BH}}/13$ is a consequence of complementarity, not an assumption. The full information-theoretic Page curve requires the island mechanism on the dynamical-metric path integral (a distinct construction); the present theorem computes the algebraic-sector saddle exchange.

### 9.1. The Lagrangian decomposition on the BTZ background.

The complementarity theorem (Theorem 5.1) decomposes the genus-$g$ bar complex into Koszul-dual halves. We evaluate this decomposition on the BTZ Maurer–Cartan element $\alpha_{\mathrm{BTZ}} \in \mathfrak{g}^{\mathrm{mod}}$ (Theorem 8.2), producing a two-saddle partition function parametrised by the black hole mass.

**Construction 9.1** (Lorentzian BTZ genus expansion). Let $\alpha_{\mathrm{BTZ}}$ be the MC element at conformal weight $h$ (non-rotating, $J = 0$). The genus-summed free energy (Proposition 8.4) is:

$$\mathcal{F}(h;\quad) = S_{\mathrm{BH}}(h)\quad^{-2} + \sum_{g\geq 1} F_g(\alpha_{\mathrm{BTZ}})\quad^{2g-2}, \tag{14}$$

where $S_{\mathrm{BH}}(h) = 2\pi\sqrt{ch/6}$ and $\quad = S_{\mathrm{BH}}^{-1}$. The Lorentzian continuation replaces the Euclidean modular parameter $\tau = i\beta/(2\pi\ell)$ by $\tau \mapsto t/(2\pi\ell) + i\epsilon$, where $t$ is Lorentzian time and $\epsilon \to 0^+$ selects the retarded contour. For an evaporating black hole, the conformal weight is time-dependent: $h(t) = h_0 - \dot{h}\,t$ with $\dot{h} > 0$, so that $S_{\mathrm{BH}}(t) = 2\pi\sqrt{c\,h(t)/6}$ decreases monotonically.

**Theorem 9.2** (Lagrangian decomposition on the BTZ background). *The complementarity decomposition (Theorem 5.1, Theorem C) applied to the genus expansion (14) produces:*

$$\mathcal{F}(h;\quad) = \mathcal{F}_+(\kappa;\, h;\quad) + \mathcal{F}_-(\kappa';\, h;\quad),$$

*where the two Lagrangian contributions are:*

$$\mathcal{F}_+(\kappa;\, h;\quad) = \kappa\quad^{-2} + \sum_{g\geq 1} \kappa\,\lambda_g^{\mathrm{FP}}\quad^{2g-2}, \qquad \kappa = c/2,$$

$$\mathcal{F}_-(\kappa';\, h;\quad) = \kappa'\quad^{-2} + \sum_{g\geq 1} \kappa'\,\lambda_g^{\mathrm{FP}}\quad^{2g-2}, \qquad \kappa' = (26-c)/2.$$

*The genus-$g$ obstruction decomposes:*

$$\mathbf{C}_g(\mathrm{Vir}_c)\big|_{\alpha_{\mathrm{BTZ}}} = \mathbf{Q}_g^+(\kappa) \oplus \mathbf{Q}_g^-(\kappa'),$$

*with fiberwise differentials satisfying $(d_{\mathrm{fib}}^+)^2 = \kappa\cdot\omega_g$ and $(d_{\mathrm{fib}}^-)^2 = \kappa'\cdot\omega_g$ on the two summands.*

*Proof.* Theorem C applied to the Virasoro algebra gives the Lagrangian eigenspace decomposition of the genus-$g$ bar complex under the Koszul involution $c \mapsto 26 - c$. The involution has period 2; its two eigenvalues on the fiberwise differential $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$ are $\kappa = c/2$ and $\kappa' = (26 - c)/2$, the curvatures of the two eigenspaces. Since $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT, Proposition 7.6), each summand contributes proportionally to its curvature eigenvalue. The genus-summed partition function decomposes accordingly. The evaluation on $\alpha_{\mathrm{BTZ}}$ preserves the decomposition: the MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ is satisfied independently on each Lagrangian half because the Koszul involution commutes with the MC operator (it acts on the parameter $c$, not on the MC variable $\alpha$). □

9.2. **The radiation entropy from saddle-point competition.** The two Lagrangian halves compete as the black hole evaporates. The radiation entropy is determined by the saddle-point prescription (the gravitational path integral selects the geometry minimising the generalised entropy), and the Page curve emerges from the exchange of dominance.

**Definition 9.3** (Evaporation rate from the genus expansion)**.** The entanglement production rate of a 2d CFT at central charge $c$ and Hawking temperature $T_H$ is determined by the genus-1 free energy. In thermal time units $\beta/(2\pi)$:

$$\dot{S}_{\mathrm{rad}} \;=\; \frac{4\pi\, F_1}{\beta/(2\pi)} \;=\; 4\pi \cdot \frac{c}{48} \;=\; \frac{\pi c}{12} \;=\; \frac{c}{6} \text{ per unit } \frac{\beta}{2\pi},$$

using $F_1 = \kappa/24 = c/48$ (Theorem 7.1). The Koszul dual boundary, at central charge $26 - c$, has entanglement production rate $(26 - c)/6$.

**Theorem 9.4** (The algebraic Page curve)**.** ***Type signature.*** *Quadrant: Open. Presentation: scalar shadow on the algebraic sector (operator → scalar trace direction). Level: 5 (modular trace + saddle competition; scalar shadow output, not operator-level reconstruction). Hypothesis package: BTZ / Cardy (BC1)–(BC3); semiclassical $S_{\mathrm{BH}} \gg 1$; Lagrangian decomposition under the Koszul involution (Theorem 9.2); island mechanism upstream (Penington, AEMM) supplies the information-theoretic content; the algebraic sector supplies the saddle-exchange shadow only.*

*The radiation entropy of an evaporating BTZ black hole is*

$$S_{\mathrm{rad}}(t) \;=\; \min\bigl(S_+(t),\;\; S_{\mathrm{BH}}(0) - S_-(t)\bigr),$$

*where:*

- $S_+(t) = (c/6)\,t$ *is the no-island contribution (entanglement growth at the boundary rate* $\dot{S}_+ = c/6$*, controlled by* $\kappa = c/2$*);*
- $S_-(t) = ((26 - c)/6)\,t$ *is the island contribution (entanglement growth at the dual rate* $\dot{S}_- = (26 - c)/6$*, controlled by* $\kappa' = (26 - c)/2$*);*
- $S_{\mathrm{BH}}(0)$ *is the initial Bekenstein–Hawking entropy.*

*The following are consequences:*

(i) ***Page time.*** *The crossing condition* $S_+(t_P) = S_{\mathrm{BH}}(0) - S_-(t_P)$ *gives:*

$$t_P \;=\; \frac{3\,S_{\mathrm{BH}}(0)}{K} \;=\; \frac{3\,S_{\mathrm{BH}}(0)}{13}, \tag{15}$$

*independent of c.*

(ii) ***Page entropy.*** $S_{\mathrm{Page}} = S_+(t_P) = \dfrac{c\,S_{\mathrm{BH}}(0)}{26}$.

(iii) ***Hawking phase*** *($t < t_P$): the boundary saddle* $\bar{B}(\mathrm{Vir}_c)$ *with curvature* $\kappa = c/2$ *dominates; radiation entropy grows at rate* $c/6$*.*

(iv) ***Island phase*** *($t > t_P$): the Koszul dual saddle* $\bar{B}(\mathrm{Vir}_{26-c})$ *with* $\kappa' = (26 - c)/2$ *dominates; radiation entropy decreases.*

(v) ***Self-dual point*** *($c = 13$): $\kappa = \kappa' = 13/2$, the two saddles have equal weight at all times, $S_{\text{Page}} = S_{\text{BH}}/2$.*

*Proof.* The proof has three steps: the Lagrangian decomposition of the replica partition function, the identification of the radiation entropy from saddle dominance, and the computation of the crossing time.

*Step 1: The replica partition function.* The radiation entropy is computed from the replica partition function $Z(n)$ on the $n$-sheeted BTZ geometry: $S_{\text{rad}} = \lim_{n\to 1}(1-n)^{-1}\log(Z(n)/Z(1)^n)$. The Lagrangian decomposition (Theorem 9.2) gives $Z(n) = Z_+(n) + Z_-(n)$, corresponding to the boundary and dual saddles. The $n$-sheeted replica of the BTZ geometry is itself a BTZ geometry with conical excess at the replica branch point; Theorem 8.2 identifies it as an MC element, so the genus expansion retains the same two-saddle decomposition at every replica number.

*Step 2: The* min *formula from the generalised entropy.* In the semiclassical regime $S_{\text{BH}} \gg 1$, the gravitational path integral is dominated by the saddle with smallest generalised entropy (the quantum extremal surface prescription). Two saddles contribute:

- The *no-island saddle*: the boundary-side partition function $Z_+$, contributing $S_{\text{rad}} = S_+(t)$ to the radiation entropy. The rate $\dot{S}_+ = c/6$ is the genus-1 entanglement rate (Definition 9.3).
- The *island saddle*: the Koszul dual partition function $Z_-$, contributing $S_{\text{rad}} = S_{\text{BH}}(0) - S_-(t)$. The subtraction arises because the island saddle computes the complement: the radiation entropy equals the total entropy minus the interior entropy, and the interior is controlled by $\kappa' = (26-c)/2$.

The dominant saddle is whichever gives the *smaller* radiation entropy: $S_{\text{rad}} = \min(S_+,\ S_{\text{BH}}(0) - S_-)$.

*Step 3: The crossing time.* The crossing condition $S_+(t_P) = S_{\text{BH}}(0) - S_-(t_P)$ reads:

$$\frac{c}{6}\,t_P \;=\; S_{\text{BH}}(0) - \frac{26-c}{6}\,t_P.$$

Adding the two rates: $t_P\big(\frac{c}{6} + \frac{26-c}{6}\big) = t_P \cdot \frac{26}{6} = S_{\text{BH}}(0)$. Solving:

$$t_P \;=\; \frac{6\,S_{\text{BH}}(0)}{26} \;=\; \frac{3\,S_{\text{BH}}(0)}{13} \;=\; \frac{3\,S_{\text{BH}}(0)}{K}.$$

The $c$-independence is exact: the rates $c/6$ and $(26-c)/6$ are individually $c$-dependent, but their sum $26/6 = \varkappa/3$ depends only on the scalar complementarity invariant $\varkappa := \kappa + \kappa' = 13$ (distinct from the Trinity ghost-charge conductor $K(\text{Vir}) := c + c^! = 26$; related by $\varkappa = \varrho_{\text{Vir}} \cdot K$, $\varrho_{\text{Vir}} = 1/2$). □

*Remark* 9.5 (The entanglement rate is not an external input). The Calabrese–Cardy entanglement rate $c/6$ is not imported from an independent argument. It is *derived* from the bar complex: the genus-1 free energy $F_1 = \kappa/24 = c/48$ (Theorem 7.1) determines the Hawking radiation rate via $\dot{S}_{\text{rad}} = 4\pi F_1/(\beta/(2\pi)) = c/6$ (Definition 9.3). The rate for the Koszul dual is $(26-c)/6$, and their sum $26/6 = K/3$ is the quantity controlling the Page time. The bar complex provides both the two saddles and their relative growth rates; the entire Page curve is internal to the genus expansion.

*Remark* 9.6 ($c$-independence of the Page time). The coefficient $3/13$ in $t_P = 3S_{\text{BH}}/13$ is purely algebraic: 3 is the entanglement normalisation ($c/6$ rather than $c/2$; equivalently, $\kappa/3$ rather than $\kappa$), and 13 is the Virasoro complementarity constant $K = 13$. The dependence on the specific black hole enters only through $S_{\text{BH}}$. For any Koszul pair with complementarity constant $K$, the Page time is $t_P = 3S_{\text{BH}}/K$.

9.3. **The island as Koszul dual dominance.** The island of the quantum extremal surface prescription has a precise algebraic incarnation: it is the locus in the genus expansion where the Koszul dual saddle dominates. The complementarity theorem fixes the number of saddles (exactly two), their curvatures ($\kappa$ and $\kappa'$), and the transition time ($t_P$).

**Theorem 9.7** (Complementarity decomposition). *At each genus $g \geq 1$, the genus-$g$ obstruction decomposes under the Koszul involution:*

$$\mathbf{C}_g(\mathrm{Vir}_c) \;=\; \mathbf{Q}_g(\mathrm{Vir}_c) \;\oplus\; \mathbf{Q}_g(\mathrm{Vir}_{26-c}).$$

*The curvature of the first summand is $\kappa = c/2$; the curvature of the second is $\kappa' = (26-c)/2$. There is no third saddle: the Koszul involution has period 2, so the pair $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$ exhausts the decomposition.*

*The transition is sharp: at $t = t_P$, the dominance switches at every genus simultaneously because the ratio $\kappa/\kappa' = c/(26-c)$ is genus-independent (the $\lambda_g^{\mathrm{FP}}$ factors cancel in the ratio of Boltzmann weights).*

*Proof.* The Lagrangian eigenspace decomposition is Theorem C applied to the Virasoro algebra (Theorem 9.2). The curvatures $\kappa$ and $\kappa'$ are the eigenvalues of $d^2_{\mathrm{fib}}/\omega_g$ on the two summands. The ratio of Boltzmann weights at genus $g$ is

$$\frac{Z_+^{(g)}}{Z_-^{(g)}} \;=\; \exp\!\Big((\kappa-\kappa')\,\lambda_g^{\mathrm{FP}} \quad {}^{2g-2}\Big) \;=\; \exp\!\Big((c-13)\,\lambda_g^{\mathrm{FP}} \quad {}^{2g-2}\Big),$$

which depends on $g$ through the common factor $\lambda_g^{\mathrm{FP}} \quad {}^{2g-2}$ but whose sign ($c > 13$ vs $c < 13$) is genus-independent. At $c = 13$: the ratio is 1 at all genera and all times. □

*Remark* 9.8 (Comparison with Penington and AEMM). The gravitational island of Penington [6] and Almheiri–Engelhardt–Marolf–Maxfield [7, 8] is the locus where the quantum extremal surface dominates the gravitational path integral. The Koszul dual saddle $\bar{B}(\mathrm{Vir}_{26-c})$ is the algebraic counterpart: the two-saddle structure, the exchange of dominance, and the $c$-independent Page time all emerge from the complementarity constant $K = 13$ alone. The identification:

| **QES / replica wormhole** | **Bar complex** |
|---|---|
| no-island saddle | $\mathbf{Q}_g(\mathrm{Vir}_c)$ |
| island saddle | $\mathbf{Q}_g(\mathrm{Vir}_{26-c})$ |
| quantum extremal surface | Koszul involution fixed locus |
| island region | $\kappa'$-eigenspace dominance |
| Page time | saddle crossing |
| two saddles, no more | period-2 involution |

The number of saddles (exactly two) is determined by the Koszul involution, not by a choice of replica geometry. The bar complex provides the mechanism; the Penington–AEMM programme provides the physical context and the fine-grained interpretation.

9.4. **The transgression algebra and the Page transition.** The transgression algebra $B_\Theta$ provides a second, independent characterisation of the Page transition. The degeneration of $B_\Theta$ at $u = \eta^2 = 0$ is the algebraic mechanism underlying the saddle exchange: the Morita-invisible Hawking phase degenerates into the information-carrying island phase.

**Proposition 9.9** (The transgression algebra at the BTZ saddle). *Let $B_\Theta$ be the transgression algebra of the bar complex $\bar{B}(\mathrm{Vir}_c)$ with MC element $\Theta = \Theta_{\mathrm{Vir}_c}$ (Section 4). Set $u := \eta^2$ where $\eta$ is the degree-one generator with $d\eta = \Theta$. At the BTZ saddle:*

(i) *For $u \neq 0$ (the generic locus): the genus-$g$ Clifford completion satisfies $\mathfrak{G}_g(B_\Theta)[u^{-1}] \cong$* $\mathrm{Mat}_{2^g}(B_\Theta[u^{-1}])$. *Higher genus is Morita-invisible: the matrix algebra carries no new invariants beyond the genus-0 data. This is the* ***Hawking phase****: the boundary saddle dominates, and the black hole appears thermal.*

(ii) *For $u = 0$ (the degenerate locus): $\mathfrak{G}_g(B_\Theta)/(\eta^2) \cong B/(\Theta) \otimes \Lambda^{2g+1}$, the exterior algebra degeneration. The $\binom{2g+1}{k}$ generators in $\Lambda^k$ resolve the Morita-invisible matrix algebra into topological sectors* *of the genus-$g$ partition function. This is the* ***island phase****: the Koszul dual saddle dominates, and the radiation carries fine-grained information.*

(iii) *The* ***Page transition*** *is the passage from $u \neq 0$ to $u = 0$. The secondary anomaly $u = \eta^2$ vanishes when the two Lagrangian saddles reach equal weight. The locus $u = 0$ is the codimension-one wall in the space of MC deformations where the matrix algebra degenerates to the exterior algebra.*

*Proof.* Parts (i) and (ii) are the two regimes of the Clifford completion. The Clifford algebra $\mathrm{Cl}_{2g+1}(u)$ is the matrix algebra $\mathrm{Mat}_{2^g}$ for $u \neq 0$ (non-degenerate quadratic form) and the exterior algebra $\Lambda^{2g+1}$ for $u = 0$ (zero form).

Part (iii): the BTZ MC element $\alpha_{\mathrm{BTZ}} = h\, e_h$ determines the curvature $\Theta|_\alpha = \kappa \cdot \omega_1$ at genus 1. As the black hole evaporates, $h(t) \to 0$ and $\alpha_{\mathrm{BTZ}}(t) \to 0$; correspondingly, $u(t) = \eta^2|_{\alpha(t)} \to 0$. The Page time $t_P$ is the time at which $u(t_P) = 0$ to leading semiclassical order, which coincides with the saddle crossing (15) because $u = 0$ is the locus where neither Lagrangian half dominates. $\square$

*Remark* 9.10 (Morita invisibility is thermality). The Morita equivalence $\mathfrak{G}_g \cong \mathrm{Mat}_{2^g}(B_\Theta[\eta^{-2}])$ for $u \neq 0$ is the algebraic expression of Hawking's argument: a thermal state carries no fine-grained information regardless of the number of replicas ($g$). The breakdown at $u = 0$ is the algebraic expression of unitarity restoration: the exterior algebra $\Lambda^{2g+1}$ resolves the thermal degeneracy. The Page curve interpolates between these two regimes; the passage is controlled by the scalar complementarity $\varkappa := \kappa + \kappa' = 13$ (the Trinity $K(\mathrm{Vir}) = 26$; $\varkappa = K/2$).

9.5. **The Borel singularity and Stokes phenomenon.** Conditionally on the genus-expansion model, the shadow free energies $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ *converge* (Bernoulli decay: $\lambda_g^{\mathrm{FP}} \sim (2\pi)^{-2g}$). The full gravitational genus expansion, including the field-sector contributions from the infinite $A_\infty$ tower (class $\mathbf{M}$), diverges: the field-dependent terms $m_k|_T$ grow as $C^k \cdot k!$ (Gevrey-1, Remark 4.7). The Page transition has a third incarnation in this divergent structure: the Stokes phenomenon of the Borel-summed genus expansion.

**Theorem 9.11** (Borel singularity from the Koszul dual). *The full gravitational genus expansion*

$$Z_{\mathrm{grav}}(\quad) = \exp\left(\sum_{g \geq 0} F_g^{\mathrm{full}} \quad {}^{2g-2}\right),$$

*where $F_g^{\mathrm{full}}$ includes both the convergent scalar shadow $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT) and the Gevrey-1 field-sector corrections, satisfies:*

(i) *The series diverges factorially: $|F_g^{\mathrm{full}}| \sim C^g \cdot (2g)!$, forced by class* $\mathbf{M}$ *(all $m_k \neq 0$).*

(ii) *The Borel transform $\hat{Z}(\zeta) = \sum_{g \geq 0} F_g^{\mathrm{full}}\, \zeta^{2g-2}/(2g-2)!$ has a singularity at the instanton action*

$$A_{\mathrm{inst}} = \frac{|\kappa' - \kappa|}{K} \cdot (2\pi)^2 = \frac{|13 - c|}{13} \cdot 4\pi^2.$$

*The singularity corresponds to the Koszul dual saddle $\bar{B}(\mathrm{Vir}_{26-c})$.*

(iii) *At $c = 13$ (self-dual): $A_{\mathrm{inst}} = 0$. The Borel singularity sits at the origin, the two saddles have equal action, and the Stokes phenomenon is maximally strong.*

(iv) *The Stokes wall of the Borel summation is the Page time: the exponentially suppressed correction $\sim e^{-A_{\mathrm{inst}}/\quad}$ becomes of order unity at $t = t_P$ and dominates thereafter.*

*Proof.* Part (i): the shadow depth $r_{\max} = \infty$ (all $m_k \neq 0$, Theorem 2.10) forces the field-sector contributions to grow as $\|m_k|_T\| \sim C^k \cdot k!$ by the Virasoro OPE recursion. The cumulative genus-$g$ free energy includes the sum over stable graph topologies at genus $g$ with vertices decorated by $\{m_k\}$; the factorial growth of $m_k$ combined with the graph-counting factor produces $|F_g^{\mathrm{full}}| \sim C^g \cdot (2g)!$ (Gevrey-1). The scalar shadow $\kappa \cdot \lambda_g^{\mathrm{FP}} \sim \kappa \cdot (2\pi)^{-2g}$ converges (Bernoulli decay) but the field sector dominates for $g \gg 1$.

Part (ii): in the resurgence framework, the Borel singularity nearest to the origin corresponds to the non-perturbative saddle closest in action. Complementarity provides exactly one such saddle: $\bar{B}(\mathrm{Vir}_{26-c})$ with curvature $\kappa'$. The tree-level free energies are $\mathcal{F}_+^{(0)} = \kappa\quad^{-2}$ and $\mathcal{F}_-^{(0)} = \kappa'\quad^{-2}$; the action difference is $|\mathcal{F}_+^{(0)} - \mathcal{F}_-^{(0)}| = |\kappa - \kappa'|\quad^{-2} = |c - 13|\quad^{-2}/1$. Normalising by the scalar instanton action $(2\pi)^2$ (the leading pole of the $\hat{A}$-genus at $x = 2\pi i$) gives $A_{\mathrm{inst}} = (2\pi)^2|\kappa' - \kappa|/K$.

Part (iii): at $c = 13$, $\kappa = \kappa'$, so $A_{\mathrm{inst}} = 0$.

Part (iv): the Stokes wall is crossed when the Borel contour passes through the singularity. On one side: $\bar{B}(\mathrm{Vir}_c)$ dominates. On the other: $\bar{B}(\mathrm{Vir}_{26-c})$ dominates. The Stokes multiplier is $\pm 1$ (the involution has period 2). The wall is crossed when the evaporating black hole reaches the critical value $\quad^{-1} = S_{\mathrm{BH}}(t_P)$; at leading order this reproduces $t_P = 3S_0/13$. □

*Remark* 9.12 (Three incarnations of the Page transition). The Page transition has three equivalent descriptions, each originating from a different structure on the bar complex:

(1) **Lagrangian:** the crossing of the two Koszul-dual saddle-point contributions to the gravitational partition function (Theorem 9.4).
(2) **Transgression:** the degeneration $u = \eta^2 \to 0$ in the transgression algebra $B_\Theta$, where the Morita-invisible matrix algebra degenerates to the exterior algebra (Proposition 9.9).
(3) **Resurgent:** the Stokes wall of the Borel-summed genus expansion, where the exponentially suppressed Koszul dual contribution becomes dominant (Theorem 9.11).

All three are controlled by the single datum $\varkappa := \kappa + \kappa' = 13$ (the scalar complementarity; the Trinity $K(\mathrm{Vir}) = c + c^! = 26$ sits upstream through $\varkappa = \varrho_{\mathrm{Vir}} \cdot K = K/2$): it determines the crossing time, the degeneration locus, and the instanton action. The number of saddles is fixed by complementarity (period-2 involution), the degeneration type by the Clifford structure (matrix vs. exterior), and the Stokes multiplier is $\pm 1$ (the involution sign).

## 10. De Sitter entropy and the gravitational Yangian

10.1. **Analytic continuation to de Sitter.** The algebraic Page curve exploits the Koszul pair $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$ at fixed signature. Analytic continuation to Lorentzian de Sitter signature ($\ell \mapsto i\ell$, equivalently $\Lambda = -1/\ell^2 \mapsto +1/\ell^2$) preserves the shadow obstruction tower and the complementarity structure.

**Theorem 10.1** (De Sitter shadow tower). *Under the Wick rotation $\ell \mapsto i\ell$ mapping* $\mathrm{AdS}_3 \to \mathrm{dS}_3$*:*

(i) *The central charge continues to $c_{\mathrm{dS}} = 3\ell_{\mathrm{dS}}/(2G)$, and the curvature to $\kappa_{\mathrm{dS}} = c_{\mathrm{dS}}/2$.*
(ii) *The Gibbons–Hawking entropy of the cosmological horizon is:*
$$S_{\mathrm{dS}} = \frac{\pi\, c_{\mathrm{dS}}}{3} = \frac{\pi\, \ell_{\mathrm{dS}}}{2G} = \frac{\mathrm{Area(cosmological\ horizon)}}{4G}.$$
(iii) *The shadow obstruction tower continues with real quantum corrections:*
$$F_g^{\mathrm{dS}} = \kappa_{\mathrm{dS}} \cdot \lambda_g^{\mathrm{FP}} = \frac{c_{\mathrm{dS}}}{2} \cdot \lambda_g^{\mathrm{FP}} \quad \textit{(UNIFORM-WEIGHT)}.$$
*In particular: $F_1^{\mathrm{dS}} = c_{\mathrm{dS}}/48$.*
(iv) *The Nariai limit $c_{\mathrm{dS}} = 13$ is the analytic continuation of the self-dual point: the de Sitter cosmological horizon entropy at the Nariai bound is self-dual under $c_{\mathrm{dS}} \mapsto 26 - c_{\mathrm{dS}}$.*

*Proof.* The Brown–Henneaux formula $c = 3\ell/(2G)$ is algebraic in $\ell$ and continues under $\ell \mapsto i\ell$ (or, equivalently, $\ell_{\mathrm{dS}} = |\ell|$). The curvature $\kappa = c/2$ is polynomial in $c$ and continues directly. The Gibbons–Hawking entropy: the cosmological horizon of $\mathrm{dS}_3$ has circumference $2\pi\ell_{\mathrm{dS}}$, so Area $= 2\pi\ell_{\mathrm{dS}}$, and $S_{\mathrm{dS}} = 2\pi\ell_{\mathrm{dS}}/(4G) = \pi\ell_{\mathrm{dS}}/(2G) = \pi c_{\mathrm{dS}}/3$. The shadow free energies $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ are polynomial in $\kappa$ and continue to real values. The Nariai limit corresponds to the maximal black hole in $\mathrm{dS}_3$, where the black hole and cosmological horizons coincide; the central charge at this point is $c_{\mathrm{dS}} = 13$, the self-dual value. □

*Remark* 10.2 (The Banks conjecture). The Banks conjecture states that the Hilbert space of de Sitter quantum gravity is finite-dimensional: $\dim(\mathcal{H}_{\mathrm{dS}}) = \exp(S_{\mathrm{dS}}) = \exp(\pi c_{\mathrm{dS}}/3)$. The convergent shadow obstruction tower (Bernoulli decay: $\lambda_g^{\mathrm{FP}} \sim (2\pi)^{-2g}$) is consistent with this finite-dimensionality: the quantum corrections to the de Sitter entropy do not produce a divergent series that would require non-perturbative completion from additional degrees of freedom.

10.2. **Horizon thermodynamics from negative curvature.** The analytic continuation $\kappa \to -|\kappa|$ has a direct interpretation in the bar complex.

**Proposition 10.3** (Negative curvature and horizon thermodynamics)**.** *The fiberwise differential at genus $g \geq 1$ satisfies $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ on $\overline{\mathcal{M}}_g$. Under $\kappa \to -|\kappa|$:*

- (i) *The curved bar complex acquires* opposite *curvature: $d_{\mathrm{fib}}^2 = -|\kappa| \cdot \omega_g$.*
- (ii) *The genus tower changes sign: $F_g^{\mathrm{dS}} = -|\kappa| \cdot \lambda_g^{\mathrm{FP}} < 0$ for $g \geq 1$.*
- (iii) *The negative free energies produce* positive *entropy contributions: $-F_g < 0$ in the exponent of $Z$ means the partition function decreases, consistent with a finite-dimensional Hilbert space.*

*The de Sitter entropy $S_{\mathrm{dS}} = \pi c_{\mathrm{dS}}/3$ is the genus-0 contribution (the classical action evaluated on the cosmological horizon saddle), with the tower of negative corrections $\{F_g^{\mathrm{dS}}\}_{g\geq 1}$ systematically reducing the effective number of microstates.*

*Proof.* The fiberwise differential is $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ (proved in Section 4). Replacing $\kappa$ by $-|\kappa|$ changes the sign of $\omega_g$. The genus-$g$ free energy $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ then becomes $F_g^{\mathrm{dS}} = -|\kappa| \cdot \lambda_g^{\mathrm{FP}} < 0$ (since $\lambda_g^{\mathrm{FP}} > 0$). □

10.3. **The gravitational Yangian at $c = 13$.** At the Koszul self-dual point $c = 13$, the Virasoro algebra acquires a distinguished algebraic structure: the ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_{13})$ carries a gravitational Yangian symmetry.

**Definition 10.4** (Gravitational Yangian)**.** The *gravitational Yangian* $\mathcal{Y}_{\mathrm{grav}}$ is the algebra of conserved charges of the shadow obstruction tower $\{S_r\}_{r\geq 2}$ at $c = 13$, acting on the ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_{13}) = T^c(s^{-1}\overline{\mathrm{Vir}}_{13})$. The shadow connection $\nabla^{\mathrm{hol}} = d - \sum_{r\geq 2} S_r$ at $c = 13$ is the generating function of the Yangian generators.

**Theorem 10.5** (Properties of the gravitational Yangian at $c = 13$)**.** *At the self-dual point $c = 13$:*

- (i) ***Self-duality.*** $\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$*: the boundary algebra equals its Koszul dual. The curvature is $\kappa = 13/2$, and $\kappa_{\mathrm{eff}} = (13 - 26)/2 = -13/2$.*
- (ii) ***Symmetric genus tower.*** $F_g(13) = F_g(26 - 13) = F_g(13)$*: the genus tower is self-dual. The free energies: $F_1 = 13/48$, $F_2 = 91/11520$.*
- (iii) ***Symmetric Page curve.*** *At $c = 13$: $\kappa = \kappa' = 13/2$, the two saddles have equal action, the Page curve is symmetric about $t_P$, and $S_{\mathrm{Page}} = S_{\mathrm{BH}}/2$.*

(iv) ***Infinite conserved charges.*** *The shadow obstructions $S_r$, $r \geq 2$, at $c = 13$ satisfy:*

$$S_2 = 13/2, \quad S_3 = -13, \quad S_4 = 10/(13 \cdot 87) = 10/1131,$$
$$S_5 = -48/(169 \cdot 87) = -48/14703.$$

*The discriminant $\Delta = 8 \cdot (13/2) \cdot 10/1131 = 40/87 \neq 0$ confirms class* **M***: the gravitational Yangian has infinitely many generators.*

(v) ***R-matrix at self-duality.*** *The classical r-matrix at $c = 13$ is $r_{13}(z) = (13/2)/z^3 + 2T/z$, and the CYBE holds by Proposition 2.7. The KZ connection (13) at $c = 13$ has self-dual monodromy: the braiding $\sigma$ satisfies $\sigma^2 = \sigma^2_{\mathrm{dual}}$ (the braiding of $\mathrm{Vir}_{13}$ and $\mathrm{Vir}^!_{13} = \mathrm{Vir}_{13}$ coincide).*

*Proof.* Parts (i)–(iii) are immediate from Theorem 5.1 and Proposition 5.2 at $c = 13$.

Part (iv): the shadow invariants at $c = 13$ are obtained by evaluating the general formulas: $S_2 = 13/2$, $S_3 = -13$, $S_4 = 10/(13(5 \cdot 13 + 22)) = 10/(13 \cdot 87) = 10/1131$, $S_5 = -48/(13^2 \cdot 87) = -48/14703$. The discriminant $\Delta = 8\kappa S_4 = 8(13/2)(10/1131) = 40/87 \neq 0$; class **M**.

Part (v): the $r$-matrix at $c = 13$ is $r_{13}(z) = (13/2)/z^3 + 2T/z$ by direct evaluation of (8). The CYBE holds for all $c$ by Proposition 2.7. Koszul self-duality of the monodromy at $c = 13$ (i.e. $\mathrm{Vir}^!_{13} = \mathrm{Vir}_{13}$, the Koszul involution $c \mapsto 26 - c$ fixing $c = 13$) implies that the Koszul dual connection is the same connection. □

*Remark* 10.6 (The role of $c = 13$ versus $c = 26$). The two distinguished central charges are:

- $c = 13$: *Koszul self-duality*. $\mathrm{Vir}^!_{13} = \mathrm{Vir}_{13}$. The complementarity constant is $K = 13$. The algebraic Page curve is symmetric. The gravitational Yangian has full self-dual structure. This is the algebraic self-duality of 3d gravity.
- $c = 26$: *matter-ghost cancellation*. $\kappa_{\mathrm{eff}} = 0$. The effective genus tower vanishes: $F^{\mathrm{eff}}_g = 0$ for all $g \geq 1$. The gravitational partition function becomes $S$-invariant. This is the bosonic string critical point, not self-duality.

The physical content of 3d gravity concentrates at $c = 13$, not $c = 26$: at $c = 13$, the gravitational perturbation series (the shadow obstruction tower) has maximal symmetry (self-duality), the algebraic Page curve has maximal symmetry (equal entropies), and the Yangian has maximal structure (self-dual braiding). At $c = 26$, the effective perturbation series vanishes.

**Conjecture 10.7** (The gravitational Yangian as symmetry algebra)**.** *At $c = 13$, the gravitational Yangian $\mathcal{Y}_{\mathrm{grav}}$ is the symmetry algebra of 3d self-dual quantum gravity in the following sense:*

(i) *The shadow obstructions $\{S_r\}_{r \geq 2}$ generate an algebra isomorphic to a quotient of the Yangian $Y(\mathfrak{vir}_{13})$ associated to the Virasoro Lie algebra at $c = 13$.*

(ii) *The shadow connection $\nabla^{\mathrm{hol}} = d - \sum_{r \geq 2} S_r$ is a connection for $\mathcal{Y}_{\mathrm{grav}}$-modules.*

(iii) *The integrability of the self-dual gravitational theory (manifested by the convergence of the scalar shadow tower) is a consequence of the Yangian symmetry.*

(iv) *The conserved charges $S_r$ at $c = 13$ satisfy the RTT-type commutation relations*

$$R_{12}(z - w)\, T_1(z)\, T_2(w) \;=\; T_2(w)\, T_1(z)\, R_{12}(z - w)$$

*with R-matrix constructed from $r_{13}(z)$.*

*Remark* 10.8 (Comparison with the holographic datum). The gravitational Yangian at $c = 13$ is the most structured specialisation of the holographic datum $\mathcal{H}(\mathrm{Vir}_c)$: at $c = 13$, all seven entries are Koszul self-dual (invariant under $c \mapsto 26 - c$), the algebraic Page curve is symmetric, the genus tower has maximal symmetry, and the Yangian structure organises the infinite $A_\infty$ tower into a representation-theoretic framework. The holographic datum at generic $c$ carries the same seven entries but without the Koszul self-duality (specific to $c = 13$) that enables the Yangian interpretation.

## 11. Gravity, topologization, and celestial holography

The algebraic-sector identification of 3d gravity is not a motivational appendix to the modular Koszul programme. The algebraic sector is the first nontrivial gravitational realisation of the homotopy-completed iterated Sugawara topologisation ladder (conditional, scope-tagged), the chiral higher-Deligne identification of the bulk, the algebraic shadow of celestial holography, and the topologization tower across the four shadow classes. None of these constructions builds the compact dynamical-metric path integral; that lift is the F12 frontier (Section 12).

**Theorem 11.1** (Conditional iterated Sugawara topologisation ladder)**.** *For a chiral algebra $\mathcal{A}$ with a tower of inner higher-spin Casimirs $\{T^{(n)}\}_{2\leq n\leq N+1}$, each admitting a BRST primitive $G^{(n)}$ with $T^{(n)} = [Q_{\mathrm{tot}}, G^{(n)}]$ on $Q$-cohomology, assume in addition that the primitives are BRST-commuting up to coherent homotopy, $[G^{(n)}, G^{(m)}] = Q$-exact, and determine independent topological directions. Then the transferred derived chiral centre carries an $E_{N+2}$-topological structure via iterated Dunn additivity. Specialisations:*

- $\mathcal{A} = \mathrm{Vir}_c$*: one inner stress tensor, cohomological and weight-completed $E_3^{\mathrm{top}}$;*
- $\mathcal{A} = \mathcal{W}_N$*: inner tower of depth $N - 1$, $E_{N+1}^{\mathrm{top}}$ under the same coherent-homotopy package;*
- $\mathcal{A} = \mathcal{W}_\infty$*: infinite depth, $E_\infty^{\mathrm{top}}$ only under convergence and completion hypotheses.*

*Consequently, 3d quantum gravity is modeled by the $N = 2$ shadow of this conditional topological tower.*

**Theorem 11.2** (Chiral higher Deligne and holographic shadow)**.** *The derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$ is acted on, in the cohomological and weight-completed sense, by $\mathsf{SC}^{\mathrm{ch,top}}$ through the heptagon edges* (3) $\leftrightarrow$ (4) $\leftrightarrow$ (5) *of the Vol II $\mathsf{SC}^{\mathrm{ch,top}}$ heptagon. On the $C_2$-cofinite non-logarithmic standard landscape, this gives the algebraic shadow of celestial holography: the celestial chiral algebra $\mathcal{A}^{\mathrm{cel}}$ of boundary soft modes, the $SC^{\mathrm{ch,top}}$ structure on $(\mathcal{A}^{\mathrm{cel}}, Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}^{\mathrm{cel}}))$, and the shadow-tower coefficients reproducing the soft-factor hierarchy. Coverage: self-dual gauge (KM), gauge+matter (DS), gravity (Virasoro + $w_{1+\infty}$).*

*Remark* 11.3 (Topologization tower: scope per class). Topologization of $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ has four regimes, with scope tags attached for every reference in this paper:

- Class G (Heisenberg): $E_3^{\mathrm{top}}$ immediate via Dunn on $F^{H_k}$ (commutative $\Rightarrow$ Dunn applies).
- Class L ($V_k(\mathfrak{g})$, $k \neq -h^\vee$): $E_3^{\mathrm{top}}$ on the *original* complex via Costello–Gwilliam factorization on $\mathbb{R} \times \mathbb{C}$ with explicit topologizing primitives $\eta_1^{(i)}, \eta_1^{(ii)}$ making $[Q, \widetilde{G}_1] = T_{\mathrm{Sug}}$ strict chain-level.
- Class C ($\beta\gamma$, $bc$): $E_3^{\mathrm{top}}$ via Friedan–Martinec–Shenker bosonisation, reducing to Heisenberg.
- Class M (Vir, $\mathcal{W}_N$): $E_3^{\mathrm{top}}$ weight-completed via half-BRST + MC5 pattern; original-complex chain-level *open* (sole remaining direction of the topologization tower theorem).
- Critical level $k = -h^\vee$: $E_2^{\mathrm{top}}$ (dimension drop, not failure).

Theorem 3.3 is the class M line of this tower, attached with the cohomological / weight-completed hypothesis package.

*Remark* 11.4 (Class M scope and logarithmic exceptions). Class M chain-level topologization on the *original* complex remains open; the weight-completed statement is proved. The scope qualifications are: (i) $\mathrm{W}(p)$ triplet tempering is conjectural pending an Adamović–Milas character-amplitude bound (Gurarie 1993 + Flohr 1996 counterexamples); (ii) non-tempered stratum is *empty* only on the non-logarithmic subset; (iii) the conditional harmonic $\beta_N$ law separates the two polynomial candidates at

$N = 4$ (13 versus 15 and 16), a parameter-question not structural; (iv) super-Yangian canonical pairing is sub-Sugawara. 3d quantum gravity with $\Lambda < 0$, the Virasoro climax at $c = 3\ell/(2G)$, and the $c$-linearity/self-duality structure are unconditional on the Virasoro Koszul locus (generic $c$); the class M chain-level conditional affects only original-complex chain-level bar/BV comparison, not the genus expansion computed in Section 4.

*Remark* 11.5 (Algebraic identification, not gravitational construction). The relationship between the Virasoro Beilinson tower and 3d gravity is an *identification of the algebraic sector*, not a construction of a compact dynamical-metric path integral. The universal Maurer–Cartan element $\Theta_{\mathrm{Vir}_c}$ of the modular convolution algebra encodes, via its MC projections to successive degrees, the complete *algebraic-sector trace* of the perturbative gravitational genus expansion. Koszul duality $\mathrm{Vir}_c \leftrightarrow \mathrm{Vir}_{26-c}$ is the algebraic shadow of S-duality; Page-time complementarity $t_P = 3S_{\mathrm{BH}}/13$ is the structural shadow of the Page transition. The full information-theoretic Page curve (Penington, AEMM) requires the island mechanism on the dynamical-metric path integral; the complementarity decomposition computes the saddle-exchange shadow on the algebraic sector. BTZ black holes are MC elements of the algebraic convolution algebra; de Sitter entropy emerges from analytic continuation $\kappa \to -|\kappa|$.

The trace direction is unconditional on the Virasoro Koszul locus. The inverse direction (algebraic sector $\to$ compact dynamical-metric path integral) is open frontier F12: it requires the BTZ / Cardy hypothesis package (modular invariance + vacuum dominance + saddle stability) plus the Maloney–Witten + Manschot–Moore Farey-tail completion, and it remains the named operator-level Pfaffian / square-root reconstruction problem (Vol I F11 + F12). The iterated Sugawara ladder (Theorem 11.1) embeds the present algebraic sector as the $N = 2$ truncation of a conditional $E_\infty$ topologisation tower; the unconditional content is the $N = 2$ algebraic identification proved here.

## 12. The F12 frontier: from algebraic sector to compact dynamical-metric path integral

The preceding sections constructed the algebraic holographic HT sector of 3d gravity at $\Lambda < 0$: the open Beilinson tower with boundary $\mathrm{Vir}_c$, bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) \simeq \mathbb{C}[[c]]$, and the $\mathsf{SC}^{\mathrm{ch,top}}$-brace structure on the pair. The trace direction (operators $\to$ scalar shadow) is unconditional on the Virasoro Koszul locus: the genus tower $F_g = \kappa\,\lambda_g^{\mathrm{FP}}$, the algebraic Page shadow $t_P = 3S_{\mathrm{BH}}/13$, the Cardy / Bekenstein–Hawking entropy, the de Sitter analytic continuation, and the gravitational Yangian at $c = 13$ are all level-5 scalar shadows of the level-4 operator algebra, computed via modular trace and clutching on the open category.

The *inverse direction*, namely the reconstruction of a compact dynamical-metric path integral from the algebraic sector, is the open frontier F12. Stating this precisely:

**Conjecture 12.1** (F12: algebraic sector $\to$ dynamical-metric path integral)**.** ***Type signature.*** *Quadrant: Open. Presentation: operator-level Pfaffian / square-root reconstruction. Level: 5 $\to$ 4 (inverse modular reconstruction; scalar-shadow to operator-algebra comparison). Hypothesis package: BTZ / Cardy (BC1)–(BC4) of Section 8; Maloney–Witten partition function $Z_{\mathrm{MW}}(\tau)$ defined via the $\mathrm{SL}_2(\mathbb{Z})/\mathbb{Z}$ Farey sum over BTZ images; Manschot–Moore mock-modular Farey-tail completion; Hall–Borcherds residual + virtual $K_0$-determinant package (Vol I F11).*

*There exists a compact holomorphic-topological saddle structure on the gravitational path integral*

$$Z_{3\mathrm{d\,QG,Vir}}(S^2 \times S^1_\tau) \;=\; q^{(1-c)/24}\,(1-q)/\eta(\tau),$$

*extending the algebraic-sector partition function of the present construction, such that:*

(i) $Z_{3\mathrm{d\,QG,Vir}}$ *equals the Maloney–Witten Farey-tail-completed sum over BTZ saddles plus thermal* $\mathrm{AdS}_3$*;*

(ii) *the inverse trace map sends the level-5 scalar tower* $\{S_r\}_{r\geq 2}$ *to the level-4 operator content of the BTZ ensemble;*

(iii) *the algebraic Page shadow* $t_P = 3S_{\mathrm{BH}}/13$ *lifts to the full information-theoretic Page curve of Penington–AEMM via the island mechanism applied to the dynamical-metric path integral.*

*Remark* 12.2 (F12 comparison problem). F12 is the conjunction of the following three constructions:

(1) *Operator-level Pfaffian* for the Virasoro family, analogous to the Igusa cusp form $\Delta_5$ on K3 × E (Vol I F11). The construction would identify the Manschot–Moore Farey-tail completion as a virtual $K_0$-determinant of an explicit operator-level object whose protected Pfaffian is the Maloney–Witten partition function.

(2) *Cross-volume bridge to Vol II / III.* Vol II's $\mathsf{SC}^{\mathrm{ch,top}}$-brace bulk is the operator-level companion of the algebraic sector identified here; the dynamical-metric path integral would emerge as the gravity-line completion of the Vol II climax. Vol III's Drinfeld double $G(X) = D(Y^+(X))$ supplies the line / brane operator data.

(3) *Resolution of the Maloney–Witten negativity problem.* The naive Maloney–Witten sum has negative Fourier coefficients in the partition function; Manschot–Moore Farey-tail completion is the candidate cure, but its compatibility with the Beilinson-tower reconstruction at level 4 is unverified. The resolution is the F12 endpoint.

Until F12 is closed, every statement of the form "the algebraic sector *is* the gravitational path integral" or "the modular form $Z_{\mathrm{3d\,QG}}$ *constructs* the BPS Hilbert space" has the wrong type: scalar shadow ≠ operator algebra. The trace direction is unconditional; the inverse direction remains the open F12 comparison problem with its own hypothesis package.

*Remark* 12.3 (The cross-volume vertical). The algebraic holographic HT sector identified here sits at levels 3–5 of the Vol I Beilinson tower:

- Level 1 (chart): $A_b = \mathrm{Vir}_c$.
- Level 2 (twisting): $\bar{B}^{\mathrm{ord}}(\mathrm{Vir}_c)$.
- Level 3 (bulk): $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c) \simeq \mathbb{C}[[c]]$, with $E_3^{\mathrm{top}}$ structure (cohomological + weight-completed scope; Class **M**).
- Level 4 (operator): BTZ MC elements, Wilson lines, $\mathrm{Vir}_{26-c}$-modules.
- Level 5 (scalar shadow): genus tower $F_g = \kappa\,\lambda_g^{\mathrm{FP}}$, Cardy entropy, algebraic Page shadow.

Vertical equivalences: level 3 of Vol I corresponds to the Vol II $\mathsf{SC}^{\mathrm{ch,top}}$-brace bulk (climax of Vol II); level 5 corresponds to the gravity-line completion (Vol II) and to the Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ (Vol III). F12 is precisely the vertical equivalence at level 4 in the gravitational direction: identifying the BTZ ensemble of the algebraic sector with the saddle ensemble of the compact dynamical-metric path integral.

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